The topological vertex as an algorithm: conventions and worked examples

Notes accompanying the Python package `topvertex`

Abstract

We consider the topological string partition function $Z(q, Q)$ of a toric Calabi-Yau threefold, which is specified by a picture: a triangulated lattice polygon. We explain how to compute it by the topological vertex of Aganagic, Klemm, Mariño and Vafa, in the combinatorial formulation of Konishi [2], the conventions of which we follow. Nothing beyond linear algebra, calculus and elementary combinatorics is assumed. Every object is defined before it is used, and every rule is stated as an explicit algorithm with a worked example. Physics and geometry appear only in separate remarks that can be skipped. In particular, we compute by hand the first terms of Z for the resolved conifold and for local $\mathbb{P}^2$, and extract the first Gopakumar-Vafa invariants $3, -6, 27, \ldots$ of the latter. We also describe a comparison with the Nekrasov partition functions of 5d linear quiver gauge theories. The notes accompany the Python package `topvertex`, which produced all the tables and all the formulas marked "generated".

Contents

1 Introduction and summary **2**

2 Lattice polygons and their triangulations **5**
2.1 Lattice points, polygons, unimodular triangles . . . 5
2.2 Unimodular triangulations . . . 6
2.3 The dual graph and its (p, q)-vectors . . . 6
2.4 Interior versus exterior . . . 8
2.5 Heights and regular triangulations . . . 8

3 Partitions and q-numbers **9**
3.1 Partitions and Young diagrams . . . 9
3.2 q-numbers . . . 10

4 Schur functions, concretely **11**
4.1 Elementary and complete symmetric functions . . . 11
4.2 Skew Schur functions by Jacobi-Trudi . . . 11
4.3 The specialization q^ρ . . . 12
4.4 The specialization $q^{\nu+\rho}$ . . . 13

5 The vertex $C_{\lambda_1\lambda_2\lambda_3}(q)$ **14**
5.1 Definition and symmetries . . . 14
5.2 Special cases . . . 14
5.3 Three computations by hand . . . 15

6 Framings and the gluing rule **15**
6.1 The framing of a flag . . . 15
6.2 Directing edges and reading off the triples . . . 18
6.3 The gluing formula . . . 19

7 Worked examples **20**
7.1 One edge with framing n and the resolved conifold . . . 20
7.2 Local $\mathbb{P}^2$ . . . 21
7.3 Local $\mathbb{P}^1 \times \mathbb{P}^1$ . . . 23

8 Free energy and Gopakumar-Vafa invariants **25**
8.1 The free energy . . . 25
8.2 The Gopakumar-Vafa expansion . . . 25
8.3 Extracting the integers . . . 26
8.4 Examples . . . 26
8.5 Tables . . . 27

9 Curve classes and truncation **28**
9.1 The class of an edge . . . 28
9.2 Truncation needs positive weights . . . 28

A Cross-check: the 5d Nekrasov partition function **29**

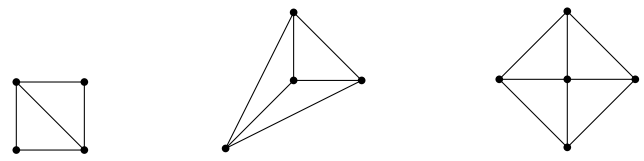
B Conventions dictionary **32**

C Answers to the exercises **34**

1 Introduction and summary

The Gopakumar-Vafa invariants n_β^g of a Calabi-Yau threefold [9] are among the most studied invariants in enumerative geometry and string theory. For compact Calabi-Yau threefolds they are very hard to compute. For toric ones, the topological vertex of Aganagic, Klemm, Mariño and Vafa [1] computes all of them, in every genus, by finite combinatorics (see [4] for a review). In these notes we consider the topological vertex as an algorithm. We follow the formulation of Konishi [2], in which the input is a lattice polygon and every intermediate quantity is an explicit rational function.

The input is a *picture*: a convex polygon in the plane with corners at integer points, cut into small triangles with corners again at integer points (the precise definitions are given in Section 2). The three pictures that we will use most are the following:



conifold local $\mathbb{P}^2$ local $\mathbb{P}^1 \times \mathbb{P}^1$

The output is a formal power series

$$Z(q;Q) = 1 + \sum_{\mathbf{d}\neq 0} Z_{\mathbf{d}}(q)\, Q^{\mathbf{d}}, \qquad Q^{\mathbf{d}} = \prod_e Q_e^{d_e}, \tag{1}$$

with one variable Q_e for every *interior* segment e of the triangulation, that is for every segment that is not on the boundary of the polygon. The coefficients $Z_{\mathbf{d}}(q)$ are *rational functions* of $q^{1/2}$. From Z one passes to the free energy $F = \log Z$ and then to a table of *integers* n_β^g, the Gopakumar-Vafa invariants, labelled by a "degree" β and a "genus" $g \geq 0$. For example, the first ones for local $\mathbb{P}^2$ are $n_1^0 = 3$, $n_2^0 = -6$, $n_3^0 = 27$, $n_3^1 = -10$. We will compute the first two of these by hand.

Physics remark (motivation only, may be skipped). A triangulated lattice polygon describes a *toric Calabi-Yau threefold* X. It is a non-compact complex manifold of complex dimension three, built by gluing copies of $\mathbb{C}^3$, one for each small triangle. Interior segments correspond to holomorphic spheres $\mathbb{P}^1 \subset X$, and $Q_e = e^{-t_e}$, where t_e (a *Kähler parameter*) is the area of that sphere. The free energy $F = \log Z$ is the generating function of the Gromov-Witten invariants, the "numbers" of holomorphic maps from Riemann surfaces of every genus into X. The Gopakumar-Vafa reorganization turns these rational numbers into integers n_β^g that count BPS states of M-theory [9]. The rule that computes Z is the *topological vertex* of [1]. It was derived from Chern-Simons theory (see [4] for a review) and proved mathematically in [11]. We follow the purely combinatorial formulation of [2], where it was used to prove that the n_β^g are integers.

The whole computation is summarized in Algorithm 0 below. Every word in it will be defined in Sections 2-9.

Algorithm 0 (The topological vertex, overview)**.**
Input: a convex lattice polygon with a regular unimodular triangulation, and a truncation degree D.
Output: the coefficients of $Z(q;Q)$ up to degree D, the free energy $F = \log Z$, and the Gopakumar-Vafa integers n_β^g up to degree D.

1. *Graph.* Put one node v_σ in each small triangle σ, one edge e_τ across each interior segment τ, and one outgoing ray across each boundary segment (Section 2.3).

2. *Framings.* For each interior segment τ compute an integer n_τ (its framing) from the four lattice points of the two triangles containing τ, after choosing a direction for e_τ (Section 6.1).

3. *Sum over partitions.* Attach a partition λ_τ (a Young diagram) to every interior segment, and add up
$$\prod_\tau (-1)^{|\lambda_\tau|(n_\tau+1)}\, q^{n_\tau \kappa(\lambda_\tau)/2}\, Q_\tau^{|\lambda_\tau|} \prod_\sigma C_{\lambda_\sigma}(q) \tag{2}$$
over all choices with $\sum_\tau w_\tau |\lambda_\tau| \le D$. Here $|\lambda|$ is the number of boxes of λ, and $\kappa(\lambda)$ is an even integer that is quadratic in the row lengths of λ. Namely, it is the N-independent part of the quadratic Casimir $\kappa(\lambda) + N|\lambda|$ of the representation of $U(N)$ labelled by λ, and its explicit form is given in (7). The function $C_{\lambda\mu\nu}(q)$ is the *topological vertex*, an explicit rational function of $q^{1/2}$ built from Schur functions (Section 5), and λ_σ is the triple of partitions around the triangle σ (Section 6.3). The positive integers w_τ are the *weights* of the interior segments. They are determined by *heights* of the triangulation, that is by numbers attached to the lattice points such that the triangulation is the projection of the lower convex hull of the lifted points (Section 2.5).

4. *Curve classes.* Replace each Q_τ by a monomial in independent variables according to the "class" of τ (Section 9).

5. *Free energy and integers.* Take $F = \log Z$ and peel off multiple covers to obtain the integers n_β^g (Section 8).

The output of Algorithm 0 can be compared with independent computations. One comparison is provided by the tables of invariants in [7, 8], some of which are reproduced in Table 3. Another one is available for the "ladder" polygons of Appendix A, for which Z is related to the Nekrasov instanton partition function of a 5d gauge theory [12, 6].

Throughout the notes, unless explicitly mentioned otherwise, we use the conventions of [2], which are also those of the package `topvertex` (file `CONVENTIONS.md`). In particular, we write
$$x = q^{1/2}, \qquad [k] = q^{k/2} - q^{-k/2} = x^k - x^{-k}, \tag{3}$$
so that rational functions of $q^{1/2}$ are rational functions of x. The vertex is the one of Definition 5.1, the framings are as in Definition 6.2, and the Gopakumar-Vafa expansion is (56). Other papers use other conventions, and Appendix B is a dictionary. Formulas and tables labelled "generated" are `\input` from the directory `notes/generated/`, which is written by the script `notes/make_tables.py` calling the package. Everything else was computed by hand and checked against the package.

The rest of the notes is organized as follows. In Section 2 we describe the input: lattice polygons, unimodular triangulations, the dual graph, and the heights of a regular triangulation. In Sections 3 and 4 we review partitions, q-numbers and Schur functions, which is everything needed for the

vertex $C_{\lambda\mu\nu}$ defined in Section 5. In Section 6 we define the framings and state the gluing rule (39) for the partition function. In Section 7 we work out the resolved conifold, local $\mathbb{P}^2$ and local $\mathbb{P}^1 \times \mathbb{P}^1$ by hand. In Section 8 we extract the Gopakumar-Vafa invariants from the free energy. Finally, in Section 9 we explain curve classes and how to truncate the sum over partitions consistently. The appendices contain the comparison with the Nekrasov partition function of a linear quiver (Appendix A), a dictionary between our conventions and those of the literature (Appendix B), and the answers to the exercises (Appendix C).

2 Lattice polygons and their triangulations

In this section we describe the input of Algorithm 0: a convex lattice polygon, a unimodular triangulation of it, the dual graph, and the heights of the triangulation.

2.1 Lattice points, polygons, unimodular triangles

Definition 2.1. A *lattice point* is a point $\omega = (a, b) \in \mathbb{R}^2$ with $a, b \in \mathbb{Z}$. A *convex lattice polygon* P is the convex hull of finitely many lattice points, not all on a line. Its *corners* are the points of P that are not in the middle of a segment contained in P, and they are lattice points. A lattice point of P is a *boundary point* if it lies on the boundary of P and an *interior point* otherwise. We write I and B for the numbers of interior and boundary lattice points.

For three points $\omega_1, \omega_2, \omega_3$ in the plane put

$$\det(\omega_1, \omega_2, \omega_3) := \det \begin{pmatrix} a_2 - a_1 & a_3 - a_1 \\ b_2 - b_1 & b_3 - b_1 \end{pmatrix}, \qquad \omega_i = (a_i, b_i). \tag{4}$$

This is *twice the signed area* of the triangle with corners $\omega_1, \omega_2, \omega_3$. It is positive exactly when the corners are listed *counterclockwise* (ccw). It is also equal to the 3×3 determinant

$$\det(\omega_1, \omega_2, \omega_3) = \det \begin{pmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ 1 & 1 & 1 \end{pmatrix}, \tag{5}$$

as one sees by subtracting the first column from the other two. We will use the column vectors $\widehat{\omega}_i = (a_i, b_i, 1)$ repeatedly.

Definition 2.2. A lattice triangle (its corners are lattice points, not on a line) is *unimodular* if $|\det(\omega_1, \omega_2, \omega_3)| = 1$, i.e. if its area is $1/2$, the smallest possible.

Theorem 2.3 (Pick)**.** *For a convex lattice polygon,* $\mathrm{Area}(P) = I + \frac{B}{2} - 1$.

Lemma 2.4. *A lattice triangle is unimodular if and only if its only lattice points are its three corners. Equivalently, the three vectors $\widehat{\omega}_1, \widehat{\omega}_2, \widehat{\omega}_3$ form a basis of $\mathbb{Z}^3$ (every integer vector is an integer combination of them).*

Proof. By the Pick theorem, the area $1/2$ means $I + B/2 = 3/2$. Since $B \geq 3$, this forces $B = 3$ and $I = 0$, and conversely. The second statement holds because an integer matrix has an integer inverse if and only if its determinant is ± 1, and the determinant is (5). □

Example 2.5. $(0,0), (1,0), (0,1)$ is unimodular ($\det = 1$, ccw). $(0,0), (2,0), (0,1)$ is not ($\det = 2$, and it contains the extra lattice point $(1,0)$).

2.2 Unimodular triangulations

Definition 2.6. A *unimodular triangulation* of a convex lattice polygon P is a finite set of unimodular lattice triangles with the following properties: (i) their union is P, (ii) two different triangles meet in a common side, a common corner, or not at all, (iii) *every* lattice point of P is a corner of some triangle. The sides of the triangles are called *segments*. A segment is *interior* if it is not contained in the boundary of P (then exactly two triangles contain it), and a *boundary segment* otherwise (then exactly one triangle contains it).

Lemma 2.7. *A unimodular triangulation of P has $T = 2\,\mathrm{Area}(P) = 2I + B - 2$ triangles, B boundary segments and $E = 3I + B - 3$ interior segments.*

Proof. Each triangle has area $1/2$, which gives T (and Pick gives the second expression). The boundary is cut into B segments by its B lattice points. Counting pairs (triangle, side) in two ways gives $3T = 2E + B$, so that $E = (3T - B)/2 = 3I + B - 3$. □

Example 2.8. The three basic examples of these notes (Figure 1):

- *Conifold*: the unit square with corners $(0,0), (1,0), (1,1), (0,1)$, cut by the diagonal from $(1,0)$ to $(0,1)$. $I = 0$, $B = 4$: two triangles, one interior segment.
- *Local* $\mathbb{P}^2$: the triangle with corners $(1,0), (0,1), (-1,-1)$ and the center $(0,0)$. Here $I = 1$, $B = 3$ and the area is $3/2$: three triangles and three interior segments (from the center to the corners). This is the unique unimodular triangulation, since all lattice points must be used.
- *Local* $\mathbb{P}^1 \times \mathbb{P}^1$ (also called local F_0): the "diamond" with corners $(1,0), (0,1), (-1,0), (0,-1)$ and center $(0,0)$. Here $I = 1$, $B = 4$: four triangles and four interior segments.

2.3 The dual graph and its (p,q)-vectors

Definition 2.9 (Dual graph, or web)**.** Given a unimodular triangulation, its *dual graph* Γ has

- one *node* v_σ for every triangle σ,
- one *internal edge* e_τ for every interior segment τ, joining the nodes of the two triangles that contain τ,
- one *external leg* (a half-infinite ray starting at v_σ) for every boundary segment of σ.

Every node has exactly three *legs* (internal edges or external legs), one through each side of its triangle.

The dual graph can be drawn with straight edges in a canonical way. For a triangle σ with corners listed counterclockwise, $(\omega_a, \omega_b, \omega_c)$, the three sides *in counterclockwise order* are $\omega_a \to \omega_b$, $\omega_b \to \omega_c$, $\omega_c \to \omega_a$. To the side $\omega_a \to \omega_b$ with direction vector $d = \omega_b - \omega_a = (d_1, d_2)$ we attach the vector

$$v = (d_2, -d_1) \qquad (d \text{ rotated by } -90^\circ). \tag{6}$$

Because the corners are ccw, v is the *outward normal* of that side. These are the (p,q)*-vectors* of the three legs at v_σ: the leg through a side is drawn in the direction of v.

Lemma 2.10 (Balancing)**.** *At every node the three (p,q)-vectors add up to zero, and consecutive ones (in ccw order) satisfy* $\det(v, v') = v_1 v_2' - v_2 v_1' = +1$. *If τ is an interior segment, the vectors attached to τ by its two triangles are opposite.*

Proof. The three side vectors of a triangle add to zero, and (6) is linear. Rotation preserves determinants, and for consecutive ccw sides $\det(\omega_b - \omega_a, \omega_c - \omega_b) = \det(\omega_a, \omega_b, \omega_c) = 1$. The two triangles containing τ traverse it in opposite directions. □

Example 2.11. For $\sigma = \{(0,0),(1,0),(0,1)\}$ (ccw in this order) the sides are $(1,0)$, $(-1,1)$, $(0,-1)$ and the legs are $v = (0,-1)$ (down), $(1,1)$ (up-right), $(-1,0)$ (left): their sum is 0.

Therefore the dual graph is a planar trivalent graph with edges perpendicular to the segments they cross. It is called the *web* or *toric diagram*. Figure 1 shows the three examples. The arrows and numbers on the internal edges will be explained in Section 6.

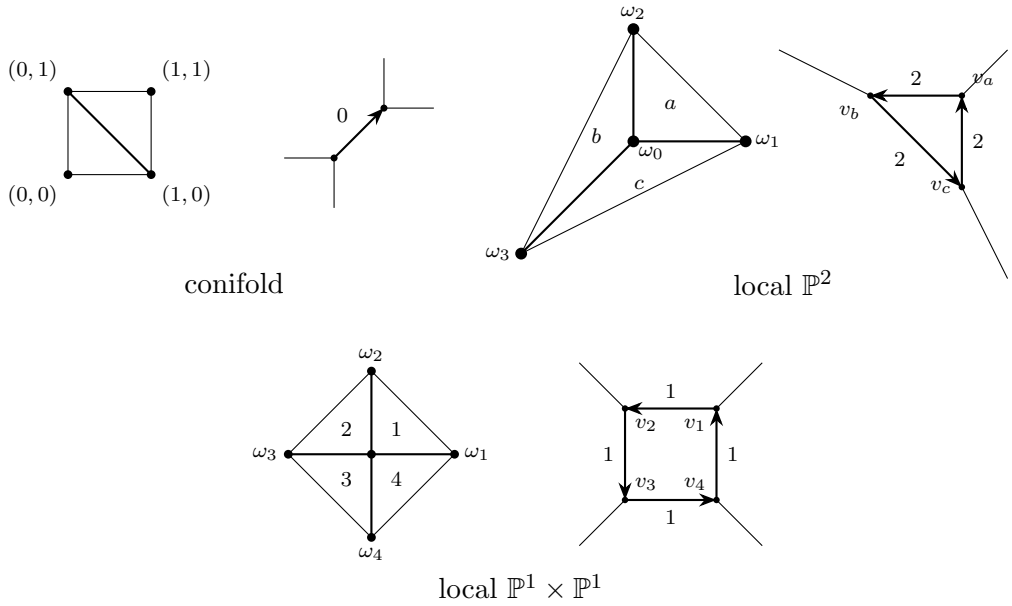


Figure 1: The three basic examples: triangulated polygon (left) and dual web (right). Interior segments are drawn thick. In the webs, the arrows are the chosen directions of the internal edges and the numbers are their framings n_e (Section 6). The external legs point in the directions of their (p,q)-vectors.

Example 2.12 (The webs of Figure 1)**.** *Local* $\mathbb{P}^2$. Label the points $\omega_0 = (0,0)$, $\omega_1 = (1,0)$, $\omega_2 = (0,1)$, $\omega_3 = (-1,-1)$ and the triangles $a = \{\omega_0,\omega_1,\omega_2\}$, $b = \{\omega_0,\omega_2,\omega_3\}$, $c = \{\omega_0,\omega_3,\omega_1\}$ (each listed ccw). The legs at v_a, in ccw order, are $(0,-1)$ (across $\omega_0\omega_1$), $(1,1)$ (external, across $\omega_1\omega_2$), $(-1,0)$ (across $\omega_2\omega_0$). At v_b: $(1,0)$ (across $\omega_0\omega_2$), $(-2,1)$ (external), $(1,-1)$ (across $\omega_3\omega_0$). At v_c: $(-1,1)$ (across $\omega_0\omega_3$), $(1,-2)$ (external), $(0,1)$ (across $\omega_1\omega_0$). Placing $v_a = (1,1)$, $v_b = (-1,1)$, $v_c = (1,-1)$ makes all internal edges point along these vectors.

Local $\mathbb{P}^1 \times \mathbb{P}^1$. With $\omega_1,\dots,\omega_4 = (1,0),(0,1),(-1,0),(0,-1)$ and triangles $k = \{\omega_0,\omega_k,\omega_{k+1}\}$ (k mod 4), the web is a square with nodes $v_1 = (1,1)$, $v_2 = (-1,1)$, $v_3 = (-1,-1)$, $v_4 = (1,-1)$ and external legs in the diagonal directions.

Remark 2.13 (A compact notation for polygons). In the package a polygon is specified by the list of its corners, written as `p=` followed by the corners `a,b` separated by underscores. For example

`p=0,0_1,0_1,1_0,1`	(conifold),
`p=1,0_0,1_-1,-1`	(local $\mathbb{P}^2$),
`p=1,0_0,1_-1,0_0,-1`	(local $\mathbb{P}^1 \times \mathbb{P}^1$).

Only the corners are listed. All the other lattice points of the polygon (interior points and boundary points between corners) are implied. Such a string is read by `topvertex.polygon.parse_polygon`.

2.4 Interior versus exterior

Only interior segments carry information in the partition function: each interior segment τ (equivalently, each internal edge e_τ) receives a formal variable Q_τ and, later, a partition λ_τ that is summed over. External legs carry nothing (formally, they carry the empty partition). Therefore the conifold has one variable, local $\mathbb{P}^2$ three and local $\mathbb{P}^1 \times \mathbb{P}^1$ four. (In Section 9 we will see that for local $\mathbb{P}^2$ the three variables must be set equal, and for $\mathbb{P}^1 \times \mathbb{P}^1$ they must be set equal in pairs.)

Physics remark (motivation only, may be skipped). The polygon is the slice of the *fan* of the toric variety X at the level one: the lattice points $\omega = (a, b)$ give the rays $\widehat{\omega} = (a, b, 1)$, and the triangles give the 3-dimensional cones, i.e. the copies of $\mathbb{C}^3$. The fact that all the rays have the third coordinate equal to 1 is the Calabi-Yau condition. Unimodularity is smoothness. The web is the image of the 1-dimensional orbits of the torus action under a moment map: internal edges are compact curves $\mathbb{P}^1$, external legs are non-compact curves $\mathbb{C}$, and interior lattice points are compact divisors (for local $\mathbb{P}^2$, the center is the surface $\mathbb{P}^2$ itself).

2.5 Heights and regular triangulations

The last piece of data that we attach to a triangulation is a set of *heights*. The heights do not enter the formula for Z. Their role is to measure the size of a term of Z, which is needed in order to truncate the series consistently (Section 9).

Definition 2.14 (Heights, regular triangulation)**.** Let $\omega_1, \dots, \omega_N$ be the lattice points of P. A *height function* h assigns an integer $h_i = h(\omega_i)$, the *height* of ω_i, to every lattice point. Lift the lattice points to the points $(\omega_i, h_i) \in \mathbb{R}^3$ and take the convex hull of the lifted points. The faces of the convex hull that are visible from below project to a subdivision of P into lattice polygons. A triangulation of P is *regular* if it is the subdivision produced in this way by some height function h. In this case $h_1, \dots, h_N$ are called *heights of the triangulation*.

Equivalently, let ϕ_h be the continuous function on P that is affine on each triangle and equal to h_i at ω_i. Then $h_1, \dots, h_N$ are heights of the triangulation if and only if ϕ_h is strictly convex across every interior segment τ. That is, for the two triangles σ and σ' that contain τ, the lifted corner of σ' opposite to τ lies strictly above the plane through the lifted triangle σ. The vertical distance between this corner and the plane is a positive integer w_τ, which we call the *weight* of τ. An explicit formula for w_τ is given in (63).

Example 2.15. For the conifold take the height 1 at $(1, 1)$ and 0 at the other three points. The lower faces of the lifted tetrahedron are the two triangles of Example 2.8, and the weight of the diagonal is 1. The heights 1 at $(1, 0)$, $(0, 1)$ and 0 at $(0, 0)$, $(1, 1)$ produce the other triangulation of the square, with the diagonal from $(0, 0)$ to $(1, 1)$. For local $\mathbb{P}^2$ and local $\mathbb{P}^1 \times \mathbb{P}^1$ one can take the height 0 at the center and 1 at the boundary points. The weights of the interior segments are then 3 and 2 respectively.

Heights of a given triangulation are not unique. For example, adding to h an affine function of the coordinates of the lattice points changes neither the triangulation nor the weights. In general, different heights give different weights, and therefore different truncations of the same series Z. The heights also provide a way to draw the dual graph. Namely, if the node v_σ is placed at the gradient of ϕ_h on σ, then the internal edge e_τ, drawn from v_σ to $v_{\sigma'}$, is w_τ times the (p, q)-vector

(6) of the corresponding leg at v_σ. In other words, the weights are the lattice lengths of the internal edges of the web.

All the triangulations that appear in these notes are regular. Non-regular unimodular triangulations exist only for larger polygons [14]. The package finds heights by linear programming (`topvertex.polygon.heights`). They can also be supplied by the user, one integer for each lattice point (the option `--heights` of the command line).

3 Partitions and q-numbers

In this section we review partitions and q-numbers and fix the notation for them.

3.1 Partitions and Young diagrams

Definition 3.1. A *partition* is a finite weakly decreasing sequence of positive integers $\lambda = (\lambda_1 \geq \lambda_2 \geq \ldots \geq \lambda_\ell > 0)$. The λ_i are its *parts*, $\ell = \ell(\lambda)$ its *length*, $|\lambda| = \sum_i \lambda_i$ its *size*. If $|\lambda| = d$ we write $\lambda \vdash d$. The *empty partition* $\varnothing = ()$ has size 0. It is convenient to set $\lambda_i = 0$ for $i > \ell(\lambda)$.

The *Young diagram* of λ is the set of *boxes* $(i, j) \in \mathbb{Z}^2$ with $1 \leq i \leq \ell(\lambda)$ and $1 \leq j \leq \lambda_i$. We draw row i (with λ_i boxes) below row $i-1$, all rows left-justified. For example $(3,1)$ is  , and $(2,2,1)$ is  . We identify a partition with its diagram and write $s \in \lambda$ for a box.

Definition 3.2. The *conjugate* (or transposed) partition λ^{t} has the transposed diagram: its j-th part is the length of the j-th column of λ, $\lambda^{\mathsf{t}}_j = \#\{i : \lambda_i \geq j\}$. For example $(3,1)^{\mathsf{t}} = (2,1,1)$. Note that $(\lambda^{\mathsf{t}})^{\mathsf{t}} = \lambda$ and $|\lambda^{\mathsf{t}}| = |\lambda|$.

Definition 3.3. The *content* of the box $s = (i, j)$ is $c(s) = j - i$ (constant along diagonals). For a partition λ put

$$\kappa(\lambda) = \sum_{i \geq 1} \lambda_i(\lambda_i - 2i + 1). \tag{7}$$

Lemma 3.4. $\kappa(\lambda) = 2\sum_{s \in \lambda} c(s)$. *Consequently* $\kappa(\lambda)$ *is even and* $\kappa(\lambda^{\mathsf{t}}) = -\kappa(\lambda)$.

Proof. The contents in row i are $1-i, 2-i, \ldots, \lambda_i - i$, with sum $\lambda_i(\lambda_i+1)/2 - i\lambda_i = \lambda_i(\lambda_i - 2i + 1)/2$. Summing over i gives the formula, and therefore evenness. Transposing exchanges i and j, so every content changes sign. □

For example, the contents of $(3,1)$ are $0, 1, 2$ in the first row and -1 in the second, so $\kappa(3,1) = 2(0 + 1 + 2 - 1) = 4$. Directly from the definition (7) one gets $3 \cdot 2 + 1 \cdot (-2) = 4$.

Physics remark (motivation only, may be skipped). The quadratic Casimir of the representation of $U(N)$ with the highest weight λ is $\sum_i \lambda_i(\lambda_i - 2i + 1 + N) = \kappa(\lambda) + N|\lambda|$, in the normalization in which the fundamental representation has the Casimir N. This is how κ enters the topological vertex: in Chern-Simons theory, changing the framing of a knot multiplies the Wilson loop in the representation λ by a phase governed by the quadratic Casimir.

Definition 3.5. For a box $s = (i, j) \in \lambda$, the *arm* $a(s) = \lambda_i - j$ is the number of boxes to its right, the *leg* $l(s) = \lambda^{\mathsf{t}}_j - i$ the number of boxes below it, and the *hook length* is $h(s) = a(s) + l(s) + 1$.

For $(3,1)$ the hook lengths are $4, 2, 1$ in the first row and 1 in the second. Transposition exchanges arms and legs, so λ and λ^{t} have the same multiset of hook lengths. Table 1 lists everything for $|\lambda| \leq 4$.

λ	diagram	λ^{t}	contents (row by row)	$\kappa(\lambda)$	hook lengths
(1)		(1)	0	0	1
(2)		$(1,1)$	$0,1$	2	$2,1$
$(1,1)$		(2)	$0 \mid -1$	-2	$2 \mid 1$
(3)		$(1,1,1)$	$0,1,2$	6	$3,2,1$
$(2,1)$		$(2,1)$	$0,1 \mid -1$	0	$3,1 \mid 1$
$(1,1,1)$		(3)	$0 \mid -1 \mid -2$	-6	$3 \mid 2 \mid 1$
(4)		$(1,1,1,1)$	$0,1,2,3$	12	$4,3,2,1$
$(3,1)$		$(2,1,1)$	$0,1,2 \mid -1$	4	$4,2,1 \mid 1$
$(2,2)$		$(2,2)$	$0,1 \mid -1,0$	0	$3,2 \mid 2,1$
$(2,1,1)$		$(3,1)$	$0,1 \mid -1 \mid -2$	-4	$4,1 \mid 2 \mid 1$
$(1,1,1,1)$		(4)	$0 \mid -1 \mid -2 \mid -3$	-12	$4 \mid 3 \mid 2 \mid 1$

Table 1: All partitions of size $1 \le |\lambda| \le 4$. Rows of the diagram are separated by "|".

3.2 q-numbers

As a reminder, $x = q^{1/2}$. All our functions will be rational functions of x, and a power $q^{k/2}$ means x^k.

Definition 3.6. For an integer k, the *q-number* is

$$[k] = q^{k/2} - q^{-k/2} = x^k - x^{-k}. \tag{8}$$

Lemma 3.7. *1.* $[0] = 0$, $[-k] = -[k]$, *and* $[k]$ *is odd under* $q \to 1/q$ *(i.e.* $x \to 1/x$*).*

2. $[k]^2 = q^k - 2 + q^{-k}$, *in particular* $[1]^2 = q - 2 + q^{-1}$.

3. For $k \ge 1$, $[k] = [1]\,(q^{(k-1)/2} + q^{(k-3)/2} + \ldots + q^{-(k-1)/2})$ *(*k *terms). In particular* $[2] = [1](q^{1/2} + q^{-1/2})$ *and* $[3] = [1](q + 1 + q^{-1})$.

4. $1 - q^{-k} = q^{-k/2}[k]$ *and* $q^k - 1 = q^{k/2}[k]$.

Proof. Direct computation. For (3) multiply out (a telescoping sum). □

Throughout the notes we use the abbreviation

$$y := [1]^2 = q - 2 + q^{-1}. \tag{9}$$

Many symmetric expressions are polynomials in y:

$$q + q^{-1} = y + 2, \qquad q - 1 + q^{-1} = y + 1, \qquad q^2 + q^{-2} = y^2 + 4y + 2,$$

$$[2]^2 = y(y+4), \qquad [3]^2 = y(y+3)^2. \quad (10)$$

(For instance $[2]^2 = [1]^2(q^{1/2}+q^{-1/2})^2 = y(q+2+q^{-1}) = y(y+4)$.) Since y itself is invariant under $q \to 1/q$, so is every polynomial in y. Conversely, every Laurent polynomial in q invariant under $q \to 1/q$ is a polynomial in y (Exercise 3.3).

Exercise 3.1. For $\lambda = (3,2)$ find λ^{t}, the contents, $\kappa(\lambda)$ (in two ways) and the hook lengths. Check that $5!/\prod_s h(s) = 5$.

Exercise 3.2. Show that $\kappa(\lambda) = \sum_i \lambda_i^2 - \sum_j (\lambda_j^{\mathsf{t}})^2$, which gives another proof of $\kappa(\lambda^{\mathsf{t}}) = -\kappa(\lambda)$.

Exercise 3.3. (a) Express $q^3 + q^{-3}$ as a polynomial in y. (b) Show by induction on the top power that every Laurent polynomial $P(q)$ with $P(q) = P(1/q)$ is a polynomial in y, and that the coefficient of the top power y^G equals the coefficient of q^G in P.

4 Schur functions, concretely

In this section we define Schur and skew Schur functions by explicit determinants, and evaluate them at the specializations that enter the vertex. Since the letter x is reserved for $q^{1/2}$, a sequence of variables will be denoted $\mathbf{u} = (u_1, u_2, u_3, \ldots)$. It may be finite or infinite.

4.1 Elementary and complete symmetric functions

Definition 4.1. The *elementary* and *complete* symmetric functions $e_k(\mathbf{u})$, $h_k(\mathbf{u})$ ($k \geq 0$) are defined by the generating functions

$$E_{\mathbf{u}}(z) = \sum_{k\geq 0} e_k(\mathbf{u}) z^k = \prod_{i\geq 1}(1+u_i z), \qquad H_{\mathbf{u}}(z) = \sum_{k\geq 0} h_k(\mathbf{u}) z^k = \prod_{i\geq 1} \frac{1}{1-u_i z}. \quad (11)$$

We also set $e_k = h_k = 0$ for $k < 0$.

In particular, $e_0 = h_0 = 1$, $e_1 = h_1 = \sum_i u_i$, $e_2 = \sum_{i<j} u_i u_j$ and $h_2 = \sum_{i\leq j} u_i u_j$. In general, e_k is the sum of all products of k *distinct* variables and h_k the sum of all monomials of degree k. Since $E_{\mathbf{u}}(-z)H_{\mathbf{u}}(z) = 1$,

$$\sum_{j=0}^{k} (-1)^j e_j(\mathbf{u}) h_{k-j}(\mathbf{u}) = 0 \qquad (k \geq 1). \quad (12)$$

Remark 4.2 (Infinite sequences). We will only use two kinds of infinite sequences, $u_i = q^{-i+1/2}$ and the same sequence with finitely many terms changed. For these, think of $q^{-1/2}$ as a small number (or as a formal variable): then all the infinite sums and products above converge (as power series in $q^{-1/2}$) and, as we show below, they are *rational functions* of $x = q^{1/2}$. From then on we only work with these rational functions, and forget the expansion.

4.2 Skew Schur functions by Jacobi-Trudi

Definition 4.3 (Jacobi-Trudi)**.** For partitions μ, ν the *skew Schur function* is the determinant

$$s_{\mu/\nu}(\mathbf{u}) = \det\left(h_{\mu_i-\nu_j-i+j}(\mathbf{u})\right)_{1\leq i,j\leq \ell(\mu)} = \det\left(e_{\mu_i^{\mathsf{t}}-\nu_j^{\mathsf{t}}-i+j}(\mathbf{u})\right)_{1\leq i,j\leq \ell(\mu^{\mathsf{t}})}. \quad (13)$$

For $\nu = \varnothing$ we write $s_\mu = s_{\mu/\varnothing}$ (the *Schur function*), and $s_\varnothing = 1$.

The equality of the two determinants is a classical theorem [13, I.(5.5)]. One can take either as the definition, and check the equality in examples. Two facts that follow from the definition ([13, I.5]): $s_{\mu/\nu} = 0$ unless $\nu \subset \mu$ (the diagram of ν is contained in that of μ), and $s_{\mu/\mu} = 1$. In particular, in all our sums over η only $\eta \subset \mu$ contributes, so the sums are finite.

Example 4.4.

$$s_{(1)} = h_1 = e_1, \qquad s_{(2)} = h_2, \qquad s_{(1,1)} = \det \begin{pmatrix} h_1 & h_2 \\ h_0 & h_1 \end{pmatrix} = h_1^2 - h_2 = e_2, \tag{14}$$

$$s_{(2,1)} = \det \begin{pmatrix} h_2 & h_3 \\ h_0 & h_1 \end{pmatrix} = h_1 h_2 - h_3, \qquad s_{(2,1)/(1)} = \det \begin{pmatrix} h_1 & h_3 \\ 0 & h_1 \end{pmatrix} = h_1^2, \tag{15}$$

$$s_{(2)/(1)} = h_1, \qquad s_{(1)/(1)} = h_0 = 1. \tag{16}$$

For $s_{(1,1)}$ the e-form gives directly e_2 (since $(1,1)^{\mathsf{t}} = (2)$), and $h_1^2 - h_2 = e_2$ is (12) for $k = 2$. For $s_{(2,1)/(1)}$: $\mu = (2,1)$, $\nu = (1,0)$, and the entries are $h_{2-1-1+1} = h_1$, $h_{2-0-1+2} = h_3$, $h_{1-1-2+1} = h_{-1} = 0$, $h_{1-0-2+2} = h_1$.

4.3 The specialization q^ρ

Definition 4.5. q^ρ denotes the infinite sequence $u_i = q^{-i+1/2}$, i.e. $q^\rho = (q^{-1/2}, q^{-3/2}, q^{-5/2}, \ldots)$. For a symmetric function f we write $f(q^\rho)$ for its value at this sequence.

Proposition 4.6.

$$h_k(q^\rho) = \frac{q^{k(k-1)/4}}{[1][2]\cdots[k]}, \qquad e_k(q^\rho) = \frac{q^{-k(k-1)/4}}{[1][2]\cdots[k]} \qquad (k \geq 0). \tag{17}$$

Proof. Put $a = q^{-1}$ and $w = q^{-1/2}z$, so that $u_i z = a^{i-1}w$. Then $E(z) = f(w) := \prod_{i\geq 0}(1 + a^i w)$ satisfies $f(w) = (1+w)f(aw)$. Writing $f(w) = \sum_k c_k w^k$ and comparing coefficients of w^k gives $c_k = a^k c_k + a^{k-1} c_{k-1}$, i.e. $c_k = \frac{a^{k-1}}{1-a^k} c_{k-1}$, $c_0 = 1$, so that

$$\prod_{i\geq 0}(1 + a^i w) = \sum_{k\geq 0} \frac{a^{k(k-1)/2}}{(1-a)(1-a^2)\cdots(1-a^k)}\, w^k. \tag{18}$$

Similarly $g(w) = \prod_{i\geq 0}(1-a^i w)^{-1}$ satisfies $(1-w)g(w) = g(aw)$, which gives $c_k = c_{k-1}/(1-a^k)$ and $g(w) = \sum_k w^k/((1-a)\cdots(1-a^k))$. Now substitute back, using $1 - q^{-j} = q^{-j/2}[j]$ (Lemma 3.7), so that $\prod_{j=1}^k (1-q^{-j}) = q^{-k(k+1)/4}[1]\cdots[k]$:

$$h_k(q^\rho) = \frac{q^{-k/2}}{\prod_{j\leq k}(1-q^{-j})} = \frac{q^{-k/2+k(k+1)/4}}{[1]\cdots[k]},$$

$$e_k(q^\rho) = \frac{q^{-k/2}q^{-k(k-1)/2}}{\prod_{j\leq k}(1-q^{-j})} = \frac{q^{-k/2-k(k-1)/2+k(k+1)/4}}{[1]\cdots[k]}, \tag{19}$$

and the exponents simplify to $k(k-1)/4$ and $-k(k-1)/4$. □

For instance $h_1(q^\rho) = e_1(q^\rho) = 1/[1]$ (which is just the geometric series $\sum_i q^{-i+1/2} = q^{-1/2}/(1-q^{-1})$), $h_2(q^\rho) = q^{1/2}/([1][2])$, $e_2(q^\rho) = q^{-1/2}/([1][2])$, $h_3(q^\rho) = q^{3/2}/([1][2][3])$.

Theorem 4.7 (Hook formula)**.**

$$s_\lambda(q^\rho) = \frac{q^{\kappa(\lambda)/4}}{\prod_{s\in\lambda}[h(s)]}. \tag{20}$$

Consequently $s_{\lambda^{\mathsf{t}}}(q^\rho) = q^{-\kappa(\lambda)/2} s_\lambda(q^\rho)$ *and* $s_\lambda(q^\rho)\, s_{\lambda^{\mathsf{t}}}(q^\rho) = \prod_{s\in\lambda}[h(s)]^{-2}$.

This is [13, I.3, Ex. 2] rewritten with q-numbers. We do not prove it, but check it in the four smallest cases, using Example 4.4, (17) and Table 1:

- $\lambda = (1)$: $s_{(1)}(q^\rho) = h_1(q^\rho) = 1/[1]$, hook formula: $q^0/[1]$. ✓
- $\lambda = (2)$: $h_2(q^\rho) = q^{1/2}/([1][2])$, hook formula: $q^{2/4}/([2][1])$. ✓
- $\lambda = (1,1)$: $e_2(q^\rho) = q^{-1/2}/([1][2])$, hook formula: $q^{-2/4}/([2][1])$. ✓
- $\lambda = (2,1)$: $h_1h_2 - h_3 = \frac{q^{1/2}}{[1]^2[2]} - \frac{q^{3/2}}{[1][2][3]} = \frac{q^{1/2}[3]-q^{3/2}[1]}{[1]^2[2][3]} = \frac{q-q^{-1}}{[1]^2[2][3]} = \frac{1}{[1]^2[3]}$, since $q^{1/2}[3] - q^{3/2}[1] = (q^2-q^{-1}) - (q^2-q) = q - q^{-1} = [2]$. Hook formula: hooks $3,1,1$ and $\kappa = 0$. ✓

The two consequences follow because λ and λ^{t} have the same hook lengths and $\kappa(\lambda^{\mathsf{t}}) = -\kappa(\lambda)$.

4.4 The specialization $q^{\nu+\rho}$

Definition 4.8. For a partition ν, $q^{\nu+\rho}$ denotes the sequence $u_i = q^{\nu_i - i + 1/2}$, $i \geq 1$ (recall $\nu_i = 0$ for $i > \ell(\nu)$). It differs from q^ρ only in the first $\ell(\nu)$ terms.

Proposition 4.9 ([2, (3)])**.**

$$\sum_{k\geq 0} h_k(q^{\nu+\rho}) z^k = \prod_{i=1}^{\ell(\nu)} \frac{1 - q^{-i+1/2}z}{1 - q^{\nu_i - i + 1/2}z} \cdot \sum_{k\geq 0} h_k(q^\rho) z^k, \tag{21}$$

$$\sum_{k\geq 0} e_k(q^{\nu+\rho}) z^k = \prod_{i=1}^{\ell(\nu)} \frac{1 + q^{\nu_i - i + 1/2}z}{1 + q^{-i+1/2}z} \cdot \sum_{k\geq 0} e_k(q^\rho) z^k. \tag{22}$$

Proof. In the product $H_{q^{\nu+\rho}}(z) = \prod_i (1-u_iz)^{-1}$ replace the first $\ell(\nu)$ factors by those of q^ρ, multiplying and dividing by the old ones. Same for E. □

The prefactor in (21) is a *finite* rational function of z, so $h_k(q^{\nu+\rho})$ is obtained by expanding it as a power series in z (only the coefficients up to z^k are needed) and multiplying by the known series $\sum h_k(q^\rho) z^k$. This, together with Jacobi-Trudi, computes every $s_{\mu/\eta}(q^{\nu+\rho})$ in finitely many steps.

Example 4.10 ($\nu = (1)$)**.** Here $q^{(1)+\rho} = (q^{1/2}, q^{-3/2}, q^{-5/2}, \ldots)$ and the prefactor is $\frac{1-az}{1-bz}$ with $a = q^{-1/2}$, $b = q^{1/2}$. Expanding, $\frac{1-az}{1-bz} = 1 + (b-a)z + (b-a)bz^2 + \ldots$ and $b - a = [1]$. Therefore

$$h_1(q^{(1)+\rho}) = h_1(q^\rho) + [1] = \frac{1}{[1]} + [1] = \frac{1+[1]^2}{[1]} = \frac{q - 1 + q^{-1}}{[1]} = \frac{y+1}{[1]}, \tag{23}$$

$$h_2(q^{(1)+\rho}) = h_2(q^\rho) + [1]h_1(q^\rho) + [1]q^{1/2} = \frac{q^{1/2}}{[1][2]} + 1 + q^{1/2}[1] = q + \frac{q^{1/2}}{[1][2]}. \tag{24}$$

(Check of the first line: $\sum_i u_i = q^{1/2} + \sum_{i\geq 2} q^{-i+1/2} = q^{1/2} + q^{-1}/[1]$, and $q^{1/2}[1] + q^{-1} = q - 1 + q^{-1}$.) Consequently, from Example 4.4,

$$s_{(1)}(q^{(1)+\rho}) = s_{(2)/(1)}(q^{(1)+\rho}) = h_1(q^{(1)+\rho}) = \frac{y+1}{[1]},$$

$$s_{(1)/(1)}(q^{(1)+\rho}) = 1, \qquad s_{(2)}(q^{(1)+\rho}) = q + \frac{q^{1/2}}{[1][2]}. \quad (25)$$

Exercise 4.1. Compute $s_{(3)}(q^\rho)$ and $s_{(1,1,1)}(q^\rho)$ from (17) and compare with the hook formula.

Exercise 4.2. Use (22) to show $e_1(q^{(1)+\rho}) = h_1(q^{(1)+\rho})$ and $s_{(1,1)}(q^{(1)+\rho}) = e_2(q^{(1)+\rho}) = q^{-1} + \frac{q^{-1/2}}{[1][2]}$.

Exercise 4.3. Check the two forms of Jacobi-Trudi (13) against each other for $\mu = (2,1)$, $\nu = \varnothing$, using (12).

5 The vertex $C_{\lambda_1\lambda_2\lambda_3}(q)$

In this section we define the vertex, describe its symmetries and special cases, and list its values for small partitions.

5.1 Definition and symmetries

Definition 5.1 (Topological vertex [2, Def. 2.2])**.** For three partitions $\lambda_1, \lambda_2, \lambda_3$,

$$C_{\lambda_1\lambda_2\lambda_3}(q) = q^{\kappa(\lambda_3)/2}\, s_{\lambda_2}(q^\rho) \sum_\eta s_{\lambda_1/\eta}\big(q^{\lambda_2^{\mathrm{t}}+\rho}\big)\, s_{\lambda_3^{\mathrm{t}}/\eta}\big(q^{\lambda_2+\rho}\big). \quad (26)$$

The sum runs over all partitions η, but only $\eta \subset \lambda_1$ and $\eta \subset \lambda_3^{\mathrm{t}}$ contribute, so it is finite. C is a rational function of $x = q^{1/2}$.

Note where the transposes are: the first skew Schur function is evaluated at $q^{\lambda_2^{\mathrm{t}}+\rho}$ and has λ_1 on top, whereas the second is evaluated at $q^{\lambda_2+\rho}$ and has λ_3^{t} on top.

Theorem 5.2 (Cyclic symmetry, [10])**.** $C_{\lambda_1\lambda_2\lambda_3} = C_{\lambda_2\lambda_3\lambda_1} = C_{\lambda_3\lambda_1\lambda_2}$.

This is not obvious from the definition (26), in which the three slots play different roles. It is proved in [10] by interpreting C as a generating function of plane partitions. We use it freely, and one can check it on the examples below. It means that only the *cyclic order* of the three partitions matters. (Exchanging two of them is *not* a symmetry. Instead one has $C_{\lambda_1\lambda_2\lambda_3} = q^{(\kappa(\lambda_1)+\kappa(\lambda_2)+\kappa(\lambda_3))/2}\, C_{\lambda_1^{\mathrm{t}}\lambda_3^{\mathrm{t}}\lambda_2^{\mathrm{t}}}$, [1, (3.12)], which we will not need, but which the package verifies, see Exercise 5.3.)

5.2 Special cases

Directly from (26), using $s_\varnothing = 1$ and $s_{\varnothing/\eta} = 0$ for $\eta \neq \varnothing$:

$$C_{\lambda\varnothing\varnothing} = s_\lambda(q^\rho), \quad (27)$$

$$C_{\lambda\mu\varnothing} = s_\mu(q^\rho)\, s_\lambda(q^{\mu^{\mathrm{t}}+\rho}), \quad (28)$$

$$C_{\lambda\varnothing\nu} = q^{\kappa(\nu)/2} \sum_\eta s_{\lambda/\eta}(q^\rho)\, s_{\nu^{\mathrm{t}}/\eta}(q^\rho), \quad (29)$$

$$C_{\varnothing\varnothing\nu} = q^{\kappa(\nu)/2} s_{\nu^{\mathrm{t}}}(q^\rho) = s_\nu(q^\rho) \quad (30)$$

(the last step by Theorem 4.7, consistent with cyclic symmetry and (27)).

5.3 Three computations by hand

Example 5.3 ($C_{(1)\varnothing\varnothing}$, $C_{(2)\varnothing\varnothing}$, $C_{(1,1)\varnothing\varnothing}$)**.** By (27) and the hook formula,

$$C_{(1)\varnothing\varnothing} = \frac{1}{[1]}, \qquad C_{(2)\varnothing\varnothing} = \frac{q^{1/2}}{[1][2]} = \frac{q^2}{(q-1)(q^2-1)}, \qquad C_{(1,1)\varnothing\varnothing} = \frac{q^{-1/2}}{[1][2]} = \frac{q}{(q-1)(q^2-1)}, \tag{31}$$

using $[1][2] = (q-1)(q^2-1)q^{-3/2}$ (Lemma 3.7(4)).

Example 5.4 ($C_{(1)(1)\varnothing}$)**.** By (28) with $\lambda = \mu = (1)$, $\mu^{\mathsf{t}} = (1)$, and Example 4.10,

$$C_{(1)(1)\varnothing} = s_{(1)}(q^{\rho})\, s_{(1)}(q^{(1)+\rho}) = \frac{1}{[1]} \cdot \frac{y+1}{[1]} = \frac{y+1}{y} = \frac{q-1+q^{-1}}{q-2+q^{-1}} = \frac{q^2-q+1}{(1-q)^2}. \tag{32}$$

By cyclic symmetry this is also $C_{(1)\varnothing(1)}$ and $C_{\varnothing(1)(1)}$. Let us check one of these directly: by (29), $C_{(1)\varnothing(1)} = q^0\big(s_{(1)}(q^{\rho})^2 + s_{(1)/(1)}(q^{\rho})^2\big) = \frac{1}{y} + 1 = \frac{y+1}{y}$, the two terms coming from $\eta = \varnothing$ and $\eta = (1)$. ✓

Example 5.5 ($C_{(1)(1)(1)}$)**.** Now $\lambda_1 = \lambda_2 = \lambda_3 = (1)$, $\kappa((1)) = 0$, and $\eta \in \{\varnothing, (1)\}$:

$$C_{(1)(1)(1)} = s_{(1)}(q^{\rho})\Big(s_{(1)}(q^{(1)+\rho})^2 + s_{(1)/(1)}(q^{(1)+\rho})^2\Big) = \frac{1}{[1]}\Big(\frac{(y+1)^2}{y} + 1\Big) = \frac{y^2+3y+1}{[1]^3}. \tag{33}$$

In terms of q: $y^2+3y+1 = q^2-q+1-q^{-1}+q^{-2}$ and $[1]^3 = q^{-3/2}(q-1)^3$, so

$$C_{(1)(1)(1)} = \frac{q^4-q^3+q^2-q+1}{q^{1/2}(q-1)^3}. \tag{34}$$

These agree with the values listed in [2, p. 6]. Table 2 lists all vertices with $|\lambda_1|+|\lambda_2|+|\lambda_3| \le 3$, one representative per cyclic orbit.

Exercise 5.1. Compute $C_{(2)(1)\varnothing} = s_{(1)}(q^{\rho})s_{(2)}(q^{(1)+\rho})$ with Example 4.10 and compare with Table 2. Then compute $C_{\varnothing(2)(1)}$ from (26) directly (one needs $s_{(1)}(q^{(2)+\rho})$) and confirm cyclic symmetry in this case.

Exercise 5.2. Show that $C_{\lambda\varnothing\varnothing}(q)$ and $C_{\lambda^{\mathsf{t}}\varnothing\varnothing}(q)$ are exchanged by $q \to 1/q$ up to the sign $(-1)^{|\lambda|}$. (In general $C_{\lambda\mu\nu}(1/q) = (-1)^{|\lambda|+|\mu|+|\nu|}C_{\lambda^{\mathsf{t}}\mu^{\mathsf{t}}\nu^{\mathsf{t}}}(q)$.)

Exercise 5.3. Verify $C_{(2)\varnothing\varnothing} = q\, C_{(1,1)\varnothing\varnothing}$, a special case of the transposition identity quoted after Theorem 5.2.

6 Framings and the gluing rule

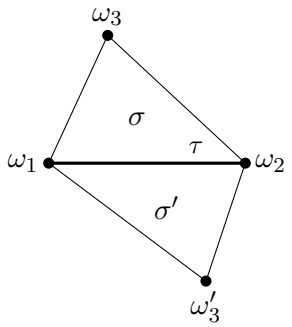
In this section we define the framings of the internal edges of the dual graph and state the gluing formula for Z.

6.1 The framing of a flag

Let τ be an interior segment with endpoints ω_1, ω_2, and let $\sigma = \{\omega_1, \omega_2, \omega_3\}$ and $\sigma' = \{\omega_1, \omega_2, \omega_3'\}$ be the two triangles containing it. A *flag* is a pair $f = (v_\sigma, e_\tau)$ of a node and an internal edge ending at it. The same edge seen from the other end is the flag $-f = (v_{\sigma'}, e_\tau)$. We always *name the endpoints of τ so that $(\omega_1, \omega_2, \omega_3)$ is counterclockwise.* Then $(\omega_2, \omega_1, \omega_3')$ is counterclockwise as well, because ω_3 and ω_3' lie on opposite sides of τ.

$(\lambda_1, \lambda_2, \lambda_3)$	$C_{\lambda_1\lambda_2\lambda_3}(q)$
$(\varnothing, \varnothing, \varnothing)$	1
$((1), \varnothing, \varnothing)$	$\dfrac{1}{[1]}$
$((2), \varnothing, \varnothing)$	$\dfrac{q^{1/2}}{[1][2]}$
$((1,1), \varnothing, \varnothing)$	$\dfrac{q^{-1/2}}{[1][2]}$
$((1), (1), \varnothing)$	$\dfrac{q - 1 + q^{-1}}{[1]^2}$
$((3), \varnothing, \varnothing)$	$\dfrac{q^{3/2}}{[1][2][3]}$
$((2,1), \varnothing, \varnothing)$	$\dfrac{1}{[1]^2[3]}$
$((1,1,1), \varnothing, \varnothing)$	$\dfrac{q^{-3/2}}{[1][2][3]}$
$((2), (1), \varnothing)$	$\dfrac{q^{5/2} - q^{3/2} + q^{-1/2}}{[1]^2[2]}$
$((1,1), (1), \varnothing)$	$\dfrac{q^{1/2} - q^{-3/2} + q^{-5/2}}{[1]^2[2]}$
$((2), \varnothing, (1))$	$\dfrac{q^{3/2} - q^{-1/2} + q^{-3/2}}{[1]^2[2]}$
$((1,1), \varnothing, (1))$	$\dfrac{q^{3/2} - q^{1/2} + q^{-3/2}}{[1]^2[2]}$
$((1), (1), (1))$	$\dfrac{q^2 - q + 1 - q^{-1} + q^{-2}}{[1]^3}$

Table 2: (Generated.) The vertex $C_{\lambda_1\lambda_2\lambda_3}(q)$ for $|\lambda_1| + |\lambda_2| + |\lambda_3| \le 3$, in lowest terms, with the denominator written as a product of q-numbers $[k] = q^{k/2} - q^{-k/2}$. One representative of each cyclic orbit is listed, and the others follow from Theorem 5.2.



Lemma 6.1. *There are unique integers a_1, a_2 with*

$$\widehat{\omega}_3' = -a_1\widehat{\omega}_1 - a_2\widehat{\omega}_2 - \widehat{\omega}_3, \qquad \text{and they satisfy} \qquad a_1 + a_2 = -2. \tag{35}$$

Proof. By Lemma 2.4, $\widehat{\omega}_1, \widehat{\omega}_2, \widehat{\omega}_3$ is a basis of $\mathbb{Z}^3$, so $\widehat{\omega}_3' = c_1\widehat{\omega}_1 + c_2\widehat{\omega}_2 + c_3\widehat{\omega}_3$ with integers c_i. Taking

the determinant with $\widehat{\omega}_1, \widehat{\omega}_2$ and using (5): $\det(\omega_1, \omega_2, \omega_3') = c_3 \det(\omega_1, \omega_2, \omega_3)$. The left side is -1 (the triangle σ' is unimodular and $(\omega_1, \omega_2, \omega_3')$ is clockwise) and the right side is $c_3 \cdot 1$, so $c_3 = -1$. Comparing third components (all equal to 1) gives $1 = c_1 + c_2 - 1$, i.e. $a_1 + a_2 = -c_1 - c_2 = -2$. □

Definition 6.2 (Framing, [2, §4.2])**.** With $(\omega_1, \omega_2, \omega_3)$ counterclockwise and a_1, a_2 as in (35), the *framing* of the flag $f = (v_\sigma, e_\tau)$ is

$$n_f = \frac{a_1 - a_2}{2} = a_1 + 1. \tag{36}$$

In plain language, the framing is an integer attached to each end of an internal edge that records how the neighborhood of the corresponding curve is twisted. In the formula for Z it appears only through the factor $q^{n\kappa(\lambda)/2}$ and a sign.

Lemma 6.3. *1.* $n_{-f} = -n_f$.

2. (Recipe.) $n_f = \det(\omega_1, \omega_3', \omega_3) - 1$, *with* det *as in (4).*

3. If an affine map $\omega \mapsto A\omega + b$ *with* $A \in SL(2, \mathbb{Z})$, $b \in \mathbb{Z}^2$, *brings* $\omega_1, \omega_2, \omega_3$ *to* $(0,0), (1,0), (0,1)$, *it brings* ω_3' *to* $(n_f + 1, -1)$.

Proof. (3) In the standard position, (35) reads $(x, y, 1) = -a_1(0,0,1) - a_2(1,0,1) - (0,1,1)$, so $\omega_3' = (-a_2, -1) = (a_1 + 2, -1) = (n_f + 1, -1)$. Affine maps with $\det A = 1$ preserve (35) (the map $\widehat{\omega} \mapsto \left(\begin{smallmatrix} A & b \\ 0 & 1 \end{smallmatrix}\right) \widehat{\omega}$ is linear) and the orientation, and therefore they preserve n_f. (2) Both sides are invariant under such maps, and in standard position $\det((0,0), (n_f + 1, -1), (0,1)) = (n_f + 1) \cdot 1 - (-1) \cdot 0 = n_f + 1$. (1) For the flag $-f$ the ccw triple is $(\omega_2, \omega_1, \omega_3')$, and (35) rewritten as $\widehat{\omega}_3 = -a_2\widehat{\omega}_2 - a_1\widehat{\omega}_1 - \widehat{\omega}_3'$ shows that the roles of a_1, a_2 are exchanged: $n_{-f} = a_2 + 1 = -1 - a_1 = -n_f$. □

Physics remark (motivation only, may be skipped). The interior segment τ is a curve $C_\tau \cong \mathbb{P}^1$ in X, and its normal bundle is $\mathcal{O}(a_1) \oplus \mathcal{O}(a_2) = \mathcal{O}(n_f - 1) \oplus \mathcal{O}(-n_f - 1)$. The condition $a_1 + a_2 = -2$ is the Calabi-Yau condition. The two flags give the two orderings of the two line bundles. For example, $n = 0$ corresponds to a "$(-1,-1)$-curve", $n = \pm 1$ to a "$(0,-2)$-curve", and $n = \pm 2$ to a "$(1,-3)$-curve". The name "framing" comes from Chern-Simons theory, where it is the choice of a normal vector field to a knot.

Example 6.4 (Framings of the basic examples)**.** We use the recipe of Lemma 6.3(2).

- *Conifold.* The segment τ goes from $(1,0)$ to $(0,1)$. In $\sigma = \{(0,0), (1,0), (0,1)\}$ the ccw order with τ first is $\omega_1 = (1,0)$, $\omega_2 = (0,1)$, $\omega_3 = (0,0)$, and $\omega_3' = (1,1)$. Therefore $n_f = \det((1,0), (1,1), (0,0)) - 1 = \det\left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right) - 1 = 0$.

- *Local* $\mathbb{P}^2$. $\tau = \omega_0\omega_1$, flag at $a = \{\omega_0, \omega_1, \omega_2\}$: $(\omega_1, \omega_2, \omega_3) \to ((0,0), (1,0), (0,1))$, $\omega_3' = (-1,-1)$, $n = \det((0,0), (-1,-1), (0,1)) - 1 = -1 - 1 = -2$. Therefore this flag has framing -2 and the flag at c has $+2$. By the rotation symmetry $(a, b) \mapsto (-b, a - b)$ of the picture (which maps $\omega_1 \to \omega_2 \to \omega_3 \to \omega_1$ and $a \to b \to c \to a$), every interior segment $\omega_0\omega_k$ has framing -2 at one end and $+2$ at the other. (In Lemma 6.3(3) form: $\omega_3' = (-1,-1) = (n+1, -1)$ gives $n = -2$.)

- *Local* $\mathbb{P}^1 \times \mathbb{P}^1$. $\tau = \omega_0\omega_1$, flag at triangle $1 = \{\omega_0, \omega_1, \omega_2\}$: $\omega_3' = \omega_4 = (0,-1) = (n+1, -1)$, so $n = -1$, and the flag at the triangle 4 has $+1$. By the $90°$ rotation symmetry all four interior segments have framings ∓ 1.

- *Local F_1.* The corners are $(1,0),(0,1),(-1,1),(0,-1)$ and the center is $\omega_0=(0,0)$. There are four triangles $\{\omega_0,\omega_k,\omega_{k+1}\}$, where $\omega_1,\dots,\omega_4$ are the points $(0,1)$, $(-1,1)$, $(0,-1)$, $(1,0)$. For the segment $\omega_0\omega_k$ at the triangle $\{\omega_0,\omega_k,\omega_{k+1}\}$ the recipe gives $n=\det(\omega_0,\omega_{k-1},\omega_{k+1})-1$ (indices $k\pm1$ taken cyclically in $\{1,2,3,4\}$, so ω_0 is always the center):

$$k=1:\ \det\big((1,0),(-1,1)\big)-1=0, \qquad k=2:\ \det\big((0,1),(0,-1)\big)-1=-1, \tag{37}$$

$$k=3:\ \det\big((-1,1),(1,0)\big)-1=-2, \qquad k=4:\ \det\big((0,-1),(0,1)\big)-1=-1, \tag{38}$$

  where $\det(u,v)$ for two vectors means $u_1v_2-u_2v_1$ (this is $\det(\omega_0,u,v)$ since $\omega_0=0$). The four framings are therefore $0,\pm1,\pm2,\pm1$: a $(-1,-1)$-curve, two $(0,-2)$-curves and a $(1,-3)$-curve (these are the exceptional curve B, two fibers F and the section $B+F$ of the Hirzebruch surface F_1, see Section 9).

- $\mathcal{O}(1)\oplus\mathcal{O}(-3)\to\mathbb{P}^1$. Points $(0,0),(0,1),(1,1),(-1,2)$, two triangles $\{(0,0),(0,1),(-1,2)\}$ and $\{(0,0),(1,1),(0,1)\}$ glued along τ from $(0,0)$ to $(0,1)$. (Their union is not convex: it is two of the three triangles of local $\mathbb{P}^2$. The gluing rule below does not need convexity.) With $\omega_1=(0,0),\omega_2=(0,1),\omega_3=(-1,2)$ (ccw) and $\omega_3'=(1,1)$: $n=\det((0,0),(1,1),(-1,2))-1=3-1=2$. The other flag has -2, and the normal bundle is $\mathcal{O}(1)\oplus\mathcal{O}(-3)$.

6.2 Directing edges and reading off the triples

Since $n_{-f}=-n_f$, the two framings of an internal edge carry the same information as one integer and a direction [2, §2.1]:

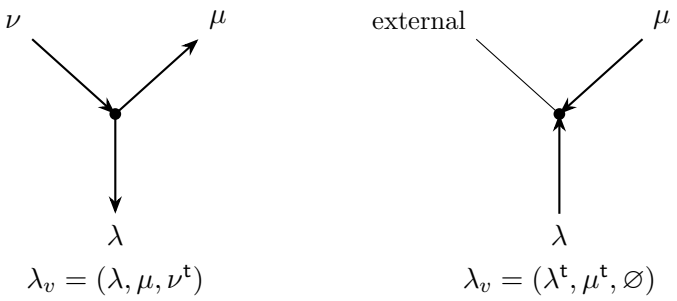
Definition 6.5. Choose (arbitrarily) a direction for every internal edge e, from its *tail* node to its *head* node. The *label* n_e of e is the framing of the flag at its tail.

Now attach a partition λ_e to every internal edge. At each node $v=v_\sigma$ we form an ordered triple of partitions λ_v, one for each of the three sides of σ:

Definition 6.6 (Triple at a node)**.** List the three sides of σ in counterclockwise order (equivalently, the three legs at v in counterclockwise order). The partition in the slot of a side is

- $\varnothing$ if the side is a boundary segment (external leg),
- λ_e if the side is crossed by the internal edge e and e points *away* from v (i.e. v is the tail of e),
- λ_e^{t} (transposed) if e points *toward* v (v is the head of e).

Which side is listed first does not matter, because the vertex is cyclically symmetric (Theorem 5.2). Only the counterclockwise cyclic order matters.



$\lambda_v=(\lambda,\mu,\nu^{\mathsf{t}})$ $\qquad$ $\lambda_v=(\lambda^{\mathsf{t}},\mu^{\mathsf{t}},\varnothing)$

Lemma 6.7 (Directions do not matter)**.** *Reversing the direction of an edge e (which replaces n_e by $-n_e$) and simultaneously renaming $\lambda_e \to \lambda_e^{\mathsf{t}}$ leaves every term of (39) below unchanged.*

Proof. The triples at the two endpoints are unchanged by construction. The edge factor becomes $(-1)^{|\lambda|(-n+1)} q^{-n\kappa(\lambda^{\mathsf{t}})/2} = (-1)^{|\lambda|(n+1)} (-1)^{-2n|\lambda|} q^{n\kappa(\lambda)/2}$, the same as before, by Lemma 3.4. □

6.3 The gluing formula

We are now ready to define the partition function.

Definition 6.8 (Partition function [2, (4), Def. 2.3])**.**

$$Z = \sum_{(\lambda_e)} \prod_e (-1)^{|\lambda_e|(n_e+1)} \, q^{n_e \kappa(\lambda_e)/2} \, Q_e^{|\lambda_e|} \prod_v C_{\lambda_v}(q), \tag{39}$$

the sum running over all assignments of a partition λ_e to each internal edge e, the first product over internal edges and the second over nodes. The coefficient $Z_{\mathbf{d}}$ of $Q^{\mathbf{d}} = \prod_e Q_e^{d_e}$ is the (finite) sum over the assignments with $|\lambda_e| = d_e$ for all e. The all-empty assignment contributes 1.

Algorithm 1 (Partition function from a triangulation)**.**
Input: lattice points $\omega_1, \dots, \omega_N$, a unimodular triangulation, positive integer weights w_e for the interior segments (all equal to 1, or the weights of Section 2.5), and a cutoff D.
Output: all coefficients $Z_{\mathbf{d}}$ with $\sum_e w_e d_e \le D$.

1. For every triangle σ, list its corners counterclockwise and create a node v_σ.
2. For every interior segment τ with triangles $\sigma \ni \omega_3$, $\sigma' \ni \omega_3'$: order its endpoints so that $(\omega_1, \omega_2, \omega_3)$ is ccw, compute $n = \det(\omega_1, \omega_3', \omega_3) - 1$, and direct the edge from v_σ to $v_{\sigma'}$ with label $n_\tau = n$. (Any other choice of direction is allowed, with $n_\tau \to -n_\tau$.)
3. For every node, record its three sides in ccw order and for each side: boundary, or (edge e, away), or (edge e, toward).
4. For every assignment (λ_e) with $\sum_e w_e |\lambda_e| \le D$:
   (a) compute the edge factor $\prod_e (-1)^{|\lambda_e|(n_e+1)} q^{n_e \kappa(\lambda_e)/2}$,
   (b) at every node, form the triple λ_v of Definition 6.6 and compute the vertex $C_{\lambda_v}(q)$ from Definition 5.1 (or Table 2),
   (c) add the product of all these to the coefficient of $\prod_e Q_e^{|\lambda_e|}$.
5. Simplify each coefficient (a rational function of $x = q^{1/2}$).

Remark 6.9. The package implements exactly this algorithm (`topvertex.partition_function`), but organizes step 4 more efficiently: it sums over the partitions of an edge as soon as both of its endpoints have been processed, which avoids enumerating all assignments at once.

Remark 6.10 (For readers who start from a web). In [1] the input is a (p, q)-web rather than a triangulation: every node carries its three leg vectors, which are *outgoing* from the node, primitive, sum to zero, and are listed counterclockwise. The framing can be read off from the web as follows. Let the internal edge e go from the node a to the node b. Let v_k be the leg at a that

immediately precedes e in counterclockwise order, and $v_{k'}$ the leg at b immediately preceding e in counterclockwise order. Then

$$n_e = v_{k'} \wedge v_k, \qquad (p_1, q_1) \wedge (p_2, q_2) := p_1 q_2 - q_1 p_2. \tag{40}$$

Proof: the leg vectors are the rotated sides (6), and rotation preserves $\wedge$. In the standard position of Lemma 6.3(3), the side of σ preceding $\tau = \omega_1\omega_2$ is $\omega_3 \to \omega_1$, i.e. $v_k = \mathrm{rot}(0,-1)$, and the side of σ' preceding τ (traversed $\omega_2 \to \omega_1$) is $\omega_3' \to \omega_2$, i.e. $v_{k'} = \mathrm{rot}(-n,1)$. Therefore $v_{k'} \wedge v_k = (-n,1) \wedge (0,-1) = n$. *Example (local $\mathbb{P}^2$, Figure 1):* the edge from v_a to v_b. At v_a the ccw legs are $(0,-1),(1,1),(-1,0)$ and the edge is $(-1,0)$, so $v_k = (1,1)$. At v_b the ccw legs are $(1,0),(-2,1),(1,-1)$ and the edge is $(1,0)$, so $v_{k'} = (1,-1)$. Then $n = (1,-1) \wedge (1,1) = 1\cdot 1 - (-1)\cdot 1 = 2$, in agreement with Example 6.4. The comparison with the way this formula is written in [1] requires care with orientation conventions, see Appendix B.

7 Worked examples

In this section we evaluate the formula (39) by hand in three basic examples: a single edge, local $\mathbb{P}^2$ and local $\mathbb{P}^1 \times \mathbb{P}^1$.

7.1 One edge with framing n and the resolved conifold

Consider a triangulation with two triangles and one interior segment, with label n (Example 6.4: the conifold has $n = 0$, the $\mathcal{O}(1) \oplus \mathcal{O}(-3)$ example has $n = 2$). Direct the edge from v_σ to $v_{\sigma'}$. The triples are $(\lambda, \varnothing, \varnothing)$ at the tail and $(\lambda^{\mathsf{t}}, \varnothing, \varnothing)$ at the head (up to cyclic order), so by (27) and Theorem 4.7

$$\begin{aligned} Z_d &= (-1)^{(n+1)d} \sum_{\lambda \vdash d} q^{n\kappa(\lambda)/2} s_\lambda(q^\rho) s_{\lambda^{\mathsf{t}}}(q^\rho) \\ &= (-1)^{(n+1)d} \sum_{\lambda \vdash d} q^{(n-1)\kappa(\lambda)/2} s_\lambda(q^\rho)^2 = (-1)^{(n+1)d} \sum_{\lambda \vdash d} \frac{q^{n\kappa(\lambda)/2}}{\prod_{s\in\lambda} [h(s)]^2}. \end{aligned} \tag{41}$$

The resolved conifold ($n = 0$). Then $Z_d = (-1)^d \sum_{\lambda\vdash d} \prod_s [h(s)]^{-2}$, and with the hook lengths of Table 1:

$$Z_1 = -\frac{1}{[1]^2}, \qquad Z_2 = \frac{1}{[2]^2[1]^2} + \frac{1}{[2]^2[1]^2} = \frac{2}{[1]^2[2]^2}, \tag{42}$$

$$Z_3 = -\Big(\frac{1}{[1]^2[2]^2[3]^2} + \frac{1}{[1]^4[3]^2} + \frac{1}{[1]^2[2]^2[3]^2}\Big) = -\frac{2y+[2]^2}{y\,[1]^2[2]^2[3]^2} = -\frac{y+6}{[1]^2[2]^2[3]^2}, \tag{43}$$

from $\lambda = (3),(2,1),(1,1,1)$ in this order and $[2]^2 = y(y+4)$. The package gives (generated):

$$\begin{aligned} Z_1 &= -\frac{1}{[1]^2} \\ Z_2 &= \frac{2}{[1]^2[2]^2} \\ Z_3 &= -\frac{q+q^{-1}+4}{[1]^2[2]^2[3]^2} = -\frac{y+6}{[1]^2[2]^2[3]^2} \end{aligned}$$

$$F_1 = -\frac{1}{[1]^2}$$
$$F_2 = -\frac{1}{2[2]^2}$$
$$F_3 = -\frac{1}{3[3]^2}$$

$$\begin{aligned} y\,f_1 &= -1 \qquad &\Longrightarrow \quad & n_1^0 = 1 \\ y\,f_2 &= 0 \qquad &\Longrightarrow \quad & n_2^g = 0 \text{ for all } g \\ y\,f_3 &= 0 \qquad &\Longrightarrow \quad & n_3^g = 0 \text{ for all } g \end{aligned}$$

The middle block is $F = \log Z$ (Section 8). It suggests the closed formula proved next. For F_2, for example: $F_2 = Z_2 - \frac{1}{2}Z_1^2 = \frac{2}{y^2(y+4)} - \frac{1}{2y^2} = \frac{-y}{2y^2(y+4)} = -\frac{1}{2[2]^2}$.

Theorem 7.1 (Cauchy identities, [13, I.4])**.** *For two sequences of variables* $\mathbf{u}, \mathbf{w}$*, as formal power series,*

$$\sum_{\lambda} s_\lambda(\mathbf{u}) s_\lambda(\mathbf{w}) = \prod_{i,j} \frac{1}{1-u_i w_j}, \qquad \sum_{\lambda} s_\lambda(\mathbf{u}) s_{\lambda^{\mathrm{t}}}(\mathbf{w}) = \prod_{i,j} (1+u_i w_j). \tag{44}$$

Proposition 7.2. *For the resolved conifold,* $Z = \prod_{k\geq 1}(1-Qq^{-k})^k$ *(expanded in* q^{-1}*) and*

$$\log Z = -\sum_{m\geq 1} \frac{Q^m}{m[m]^2}. \tag{45}$$

For a single edge with $n = 1$ *(a* $(0,-2)$*-curve),* $Z = \prod_k (1-Qq^{-k})^{-k}$ *and* $\log Z = +\sum_m \frac{Q^m}{m[m]^2}$.

Proof. By the first form of (41) with $n = 0$, $Z = \sum_\lambda s_\lambda(q^\rho) s_{\lambda^{\mathrm{t}}}(-Qq^\rho)$, where $-Qq^\rho$ is the sequence $(-Qq^{-i+1/2})_i$ (since s_μ is homogeneous of degree $|\mu|$). The second Cauchy identity gives $Z = \prod_{i,j\geq 1}(1-Qq^{-i-j+1})$. Exactly k pairs (i,j) have $i+j-1 = k$, so $Z = \prod_k (1-Qq^{-k})^k$, and

$$\log Z = \sum_k k \log(1-Qq^{-k}) = -\sum_k k \sum_m \frac{Q^m q^{-km}}{m} = -\sum_m \frac{Q^m}{m} \frac{q^{-m}}{(1-q^{-m})^2} = -\sum_m \frac{Q^m}{m[m]^2}, \tag{46}$$

using $\sum_k k t^k = t/(1-t)^2$ and $q^{-m}/(1-q^{-m})^2 = 1/(q^{m/2}-q^{-m/2})^2$. For $n = 1$ the sign is $(-1)^{2d} = 1$ and $Z = \sum_\lambda s_\lambda(q^\rho) s_\lambda(Qq^\rho)$, so that one can use the first Cauchy identity. The identities hold as power series in $q^{-1/2}$, but both sides of each Q^m-coefficient are rational functions, so they hold as rational functions. □

7.2 Local $\mathbb{P}^2$

Setup. The points are $\omega_0 = (0,0)$, $\omega_1 = (1,0)$, $\omega_2 = (0,1)$, $\omega_3 = (-1,-1)$. The triangles are $a = \{\omega_0, \omega_1, \omega_2\}$, $b = \{\omega_0, \omega_2, \omega_3\}$, $c = \{\omega_0, \omega_3, \omega_1\}$ (ccw). The internal edges e_1, e_2, e_3 cross the segments $\omega_0\omega_1, \omega_0\omega_2, \omega_0\omega_3$. By Example 6.4 each segment has framings ± 2, and we direct the edges so that all labels are $+2$:

$$e_1 :\ v_c \to v_a, \qquad e_2 :\ v_a \to v_b, \qquad e_3 :\ v_b \to v_c, \qquad n_1 = n_2 = n_3 = 2 \tag{47}$$

(Figure 1). The sides of a in ccw order are $\omega_0\omega_1$ (e_1, which points toward v_a), $\omega_1\omega_2$ (boundary), $\omega_2\omega_0$ (e_2, away from v_a). Doing the same at b and c:

$$\lambda_{v_a} = (\lambda_1^{\mathsf{t}}, \varnothing, \lambda_2), \qquad \lambda_{v_b} = (\lambda_2^{\mathsf{t}}, \varnothing, \lambda_3), \qquad \lambda_{v_c} = (\lambda_3^{\mathsf{t}}, \varnothing, \lambda_1). \tag{48}$$

The edge factors are $(-1)^{3|\lambda_i|}q^{2\kappa(\lambda_i)/2} = (-1)^{|\lambda_i|}q^{\kappa(\lambda_i)}$. All three interior segments represent the same curve class (Section 9), so we set $Q_1 = Q_2 = Q_3 = Q$:

$$Z^{\mathbb{P}^2} = \sum_{\lambda_1,\lambda_2,\lambda_3} (-Q)^{|\lambda_1|+|\lambda_2|+|\lambda_3|}\, q^{\kappa(\lambda_1)+\kappa(\lambda_2)+\kappa(\lambda_3)}\, C_{\lambda_1^{\mathsf{t}}\varnothing\lambda_2}\, C_{\lambda_2^{\mathsf{t}}\varnothing\lambda_3}\, C_{\lambda_3^{\mathsf{t}}\varnothing\lambda_1}. \tag{49}$$

We need only vertices with an empty middle slot, $C_{\mu\varnothing\nu}$, given by (29). In particular, $C_{\mu\varnothing\varnothing} = s_\mu(q^\rho)$, $C_{\varnothing\varnothing\nu} = s_\nu(q^\rho)$, and $C_{(1)\varnothing(1)} = (y+1)/y$ (Example 5.4).

Degree 1. One of the λ_i is (1), the others empty: 3 terms. For $\lambda_1 = (1)$: sign -1, q^0, $C_{\lambda_{v_a}} = C_{(1)\varnothing\varnothing} = 1/[1]$, $C_{\lambda_{v_b}} = C_{\varnothing\varnothing\varnothing} = 1$, $C_{\lambda_{v_c}} = C_{\varnothing\varnothing(1)} = 1/[1]$. The other two are the same by symmetry, so

$$Z_1 = -\frac{3}{[1]^2} = -\frac{3}{y}. \tag{50}$$

Degree 2. Either one λ_i has size 2 (3 edges $\times$ 2 partitions $= 6$ terms) or two of them are (1) (3 terms):

$(\lambda_1,\lambda_2,\lambda_3)$	sign$\cdot q^{\sum\kappa}$	$C_{\lambda_{v_a}}$	$C_{\lambda_{v_b}}$	$C_{\lambda_{v_c}}$	term
$((2),\varnothing,\varnothing)$	q^2	$C_{(1,1)\varnothing\varnothing} = \frac{q^{-1/2}}{[1][2]}$	1	$C_{\varnothing\varnothing(2)} = \frac{q^{1/2}}{[1][2]}$	$\frac{q^2}{[1]^2[2]^2}$
$((1,1),\varnothing,\varnothing)$	q^{-2}	$C_{(2)\varnothing\varnothing} = \frac{q^{1/2}}{[1][2]}$	1	$C_{\varnothing\varnothing(1,1)} = \frac{q^{-1/2}}{[1][2]}$	$\frac{q^{-2}}{[1]^2[2]^2}$
$((1),(1),\varnothing)$	1	$C_{(1)\varnothing(1)} = \frac{y+1}{y}$	$C_{(1)\varnothing\varnothing} = \frac{1}{[1]}$	$C_{\varnothing\varnothing(1)} = \frac{1}{[1]}$	$\frac{y+1}{y^2}$
and the same with $(\lambda_1,\lambda_2,\lambda_3)$ rotated: $(\varnothing,\lambda,\varnothing)$, $(\varnothing,\varnothing,\lambda)$ and $(\varnothing,(1),(1))$, $((1),\varnothing,(1))$.					

(For instance, for $(\varnothing,(1),(1))$ the vertex $C_{(1)\varnothing(1)}$ sits at v_b, and for $((1),\varnothing,(1))$ at v_c. The other two vertices are $1/[1]$ each.) Adding up, with $q^2+q^{-2} = y^2+4y+2$ and $[1]^2[2]^2 = y^2(y+4)$:

$$Z_2 = \frac{3(q^2+q^{-2})}{[1]^2[2]^2} + \frac{3(q-1+q^{-1})}{[1]^4} = \frac{3(y^2+4y+2)+3(y+1)(y+4)}{y^2(y+4)} = \frac{6y^2+27y+18}{[1]^2[2]^2}. \tag{51}$$

Degree 3. There are $3\cdot 3$ terms with one partition of size 3, $6\cdot 2$ terms of the form (size 2, size 1, empty) in some order, and one term $((1),(1),(1))$: 22 terms. The package gives (generated, the two lower blocks are explained in Section 8):

$$Z_1 = -\frac{3}{[1]^2}$$

$$Z_2 = \frac{6(q^2+q^{-2})+3(q+q^{-1})}{[1]^2[2]^2} = \frac{6y^2+27y+18}{[1]^2[2]^2}$$

$$Z_3 = -\frac{10(q^6+q^{-6})+7(q^5+q^{-5})+8(q^4+q^{-4})+3(q^3+q^{-3})+10(q^2+q^{-2})+26(q+q^{-1})+34}{[1]^2[2]^2[3]^2}$$

$$= -\frac{10y^6 + 127y^5 + 618y^4 + 1432y^3 + 1588y^2 + 756y + 162}{[1]^2[2]^2[3]^2}$$

$$\begin{aligned} F_1 &= -\frac{3}{[1]^2} \\ F_2 &= \frac{12(q+q^{-1})+21}{2[2]^2} = \frac{12y+45}{2[2]^2} \\ F_3 &= -\frac{10(q^3+q^{-3})+27(q^2+q^{-2})+54(q+q^{-1})+62}{[3]^2} \\ &= -\frac{10y^3+87y^2+252y+244}{[3]^2} \end{aligned}$$

$$\begin{aligned} y\,f_1 &= -3 && \Longrightarrow \quad n_1^0 = 3 \\ y\,f_2 &= 6 && \Longrightarrow \quad n_2^0 = -6 \\ y\,f_3 &= -10y - 27 && \Longrightarrow \quad n_3^0 = 27,\ n_3^1 = -10 \end{aligned}$$

The first two lines agree with our hand computation.

7.3 Local $\mathbb{P}^1 \times \mathbb{P}^1$

Setup. The points are $\omega_0 = (0,0)$ and $\omega_1, \ldots, \omega_4 = (1,0), (0,1), (-1,0), (0,-1)$. The triangles are $k = \{\omega_0, \omega_k, \omega_{k+1}\}$ (indices mod 4). The internal edge e_k crosses $\omega_0\omega_k$ and lies between the triangles $k-1$ and k. By Example 6.4 the framing of e_k is -1 at triangle k and $+1$ at triangle $k-1$, so we direct e_k from v_{k-1} to v_k with label $n_k = 1$ (Figure 1). At v_k the ccw sides are $\omega_0\omega_k$ (e_k, toward v_k), $\omega_k\omega_{k+1}$ (boundary) and $\omega_{k+1}\omega_0$ (e_{k+1}, away), so $\lambda_{v_k} = (\lambda_k^{\mathsf{t}}, \varnothing, \lambda_{k+1})$. The edge factors are $(-1)^{2|\lambda_k|}q^{\kappa(\lambda_k)/2} = q^{\kappa(\lambda_k)/2}$. The segments $\omega_0\omega_1$ and $\omega_0\omega_3$ represent one curve class B, and $\omega_0\omega_2$, $\omega_0\omega_4$ another class F (Section 9). Therefore we set $Q_1 = Q_3 = Q_B$, $Q_2 = Q_4 = Q_F$:

$$Z^{\mathbb{P}^1\times\mathbb{P}^1} = \sum_{\lambda_1,\ldots,\lambda_4} Q_B^{|\lambda_1|+|\lambda_3|} Q_F^{|\lambda_2|+|\lambda_4|} \prod_{k=1}^{4} q^{\kappa(\lambda_k)/2}\, C_{\lambda_k^{\mathsf{t}}\varnothing\lambda_{k+1}} \qquad (\lambda_5 = \lambda_1). \tag{52}$$

Total degree ≤ 2. We write $Z_{(d_B,d_F)}$ for the coefficient of $Q_B^{d_B}Q_F^{d_F}$.

- $(1,0)$: $\lambda_1 = (1)$ or $\lambda_3 = (1)$. For $\lambda_1 = (1)$: $C_{\lambda_{v_1}} = C_{(1)\varnothing\varnothing}$, $C_{\lambda_{v_4}} = C_{\varnothing\varnothing(1)}$, giving $1/y$. Therefore $Z_{(1,0)} = 2/y$, and likewise $Z_{(0,1)} = 2/y$.

- $(2,0)$: (i) $\lambda_1 = \mu \vdash 2$, others empty: $q^{\kappa(\mu)/2}C_{\mu^{\mathsf{t}}\varnothing\varnothing}C_{\varnothing\varnothing\mu} = q^{\kappa(\mu)/2}s_{\mu^{\mathsf{t}}}(q^\rho)s_\mu(q^\rho) = s_\mu(q^\rho)^2$, which gives $\frac{q+q^{-1}}{[1]^2[2]^2}$ summed over $\mu = (2), (1,1)$, (ii) the same for λ_3, (iii) $\lambda_1 = \lambda_3 = (1)$: the four vertices are $C_{(1)\varnothing\varnothing}, C_{\varnothing\varnothing(1)}, C_{(1)\varnothing\varnothing}, C_{\varnothing\varnothing(1)}$, giving $1/y^2$. Altogether

$$Z_{(2,0)} = \frac{2(q+q^{-1})}{[1]^2[2]^2} + \frac{1}{[1]^4} = \frac{2(y+2)+(y+4)}{y^2(y+4)} = \frac{3y+8}{[1]^2[2]^2}, \tag{53}$$

  and the same for $Z_{(0,2)}$.

- $(1,1)$: one B-edge and one F-edge carry (1). All four such pairs are adjacent (they share a node), e.g. $\lambda_1 = \lambda_2 = (1)$ gives $C_{\lambda_{v_1}} = C_{(1)\varnothing(1)} = \frac{y+1}{y}$, $C_{\lambda_{v_2}} = C_{(1)\varnothing\varnothing}$, $C_{\lambda_{v_4}} = C_{\varnothing\varnothing(1)}$, $C_{\lambda_{v_3}} = 1$. Therefore $Z_{(1,1)} = 4\,\frac{y+1}{y^2}$.

The package gives (generated):

$$Z_{(1,0)} = \frac{2}{[1]^2}$$
$$Z_{(0,1)} = \frac{2}{[1]^2}$$
$$Z_{(2,0)} = \frac{3(q+q^{-1})+2}{[1]^2[2]^2} = \frac{3y+8}{[1]^2[2]^2}$$
$$Z_{(1,1)} = \frac{4(q+q^{-1})-4}{[1]^4} = \frac{4y+4}{[1]^4}$$
$$Z_{(0,2)} = \frac{3(q+q^{-1})+2}{[1]^2[2]^2} = \frac{3y+8}{[1]^2[2]^2}$$

$$F_{(1,0)} = \frac{2}{[1]^2}$$
$$F_{(0,1)} = \frac{2}{[1]^2}$$
$$F_{(2,0)} = \frac{1}{[2]^2}$$
$$F_{(1,1)} = \frac{4}{[1]^2}$$
$$F_{(0,2)} = \frac{1}{[2]^2}$$

$$\begin{aligned}
y\,f_{(1,0)} = 2 &\quad\Longrightarrow\quad n^0_{(1,0)} = -2\\
y\,f_{(0,1)} = 2 &\quad\Longrightarrow\quad n^0_{(0,1)} = -2\\
y\,f_{(2,0)} = 0 &\quad\Longrightarrow\quad n^g_{(2,0)} = 0 \text{ for all } g\\
y\,f_{(1,1)} = 4 &\quad\Longrightarrow\quad n^0_{(1,1)} = -4\\
y\,f_{(0,2)} = 0 &\quad\Longrightarrow\quad n^g_{(0,2)} = 0 \text{ for all } g
\end{aligned}$$

Remark 7.3 (Local toric surfaces in general). The two examples above are instances of a general pattern [2, §2.5.2]. If the polygon has a single interior point ω_0 and the boundary points $\omega_1, \dots, \omega_r$ are listed counterclockwise, the triangulation is the "star" $\{\omega_0, \omega_k, \omega_{k+1}\}$ and the same reasoning gives

$$Z = \sum_{\lambda_1,\dots,\lambda_r} \prod_{k=1}^{r} (-1)^{(m_k+1)|\lambda_k|} q^{m_k \kappa(\lambda_k)/2} Q_k^{|\lambda_k|}\, C_{\lambda_k^{\mathrm{t}} \varnothing \lambda_{k+1}}, \qquad m_k = 1 - \det(\omega_0, \omega_{k-1}, \omega_{k+1}), \tag{54}$$

where e_k is directed from triangle $k-1$ to triangle k and m_k is its label. Local $\mathbb{P}^2$ has $m = (2,2,2)$, local $\mathbb{P}^1 \times \mathbb{P}^1$ has $m = (1,1,1,1)$, and local F_1 (with the points of Example 6.4) has $m = (0,1,2,1)$.

Physics remark (motivation only, may be skipped). For a star triangulation, X is the total space of the canonical bundle of a compact toric surface S, and e_k is a curve $C_k \subset S$ with self-intersection $C_k \cdot C_k = m_k - 1$: lines in $\mathbb{P}^2$ have self-intersection $+1$, the rulings of $\mathbb{P}^1 \times \mathbb{P}^1$ have 0, and F_1 has curves with -1 (the exceptional curve), 0 (fibers) and $+1$.

Exercise 7.1. Using only Z_1, Z_2, Z_3 of the conifold computed by hand above and $\log(1+W) = W - W^2/2 + W^3/3 - \ldots$, check that $F_3 = Z_3 - Z_1 Z_2 + \frac{1}{3} Z_1^3 = -\frac{1}{3[3]^2}$. (Use $[3]^2 = y(y+3)^2$.)

Exercise 7.2. For local F_1, use (54) with $m = (0,1,2,1)$ and $Q_1 = Q_B$, $Q_2 = Q_4 = Q_F$, $Q_3 = Q_B Q_F$ (Section 9) to compute the coefficients Z_B of Q_B and Z_{B+F} of $Q_B Q_F$, and $F_{B+F} = Z_{B+F} - Z_B Z_F$. (After Section 8 one will be able to read off $n_B^0 = 1$ and $n_{B+F}^0 = 3$.)

Exercise 7.3. For local $\mathbb{P}^2$, compute the contribution to Z_3 of the 9 terms in which a single λ_i has size 3.

8 Free energy and Gopakumar-Vafa invariants

In this section we define the free energy and the Gopakumar-Vafa invariants, and describe how the invariants are extracted from Z.

8.1 The free energy

After identifying the variables according to curve classes (Section 9), Z is a power series $Z = 1 + \sum_{\beta \neq 0} Z_\beta Q^\beta$ in variables $Q = (Q_1, \ldots, Q_r)$, where $\beta = (\beta_1, \ldots, \beta_r)$ runs over vectors of non-negative integers ("degrees" or "classes") and $Q^\beta = \prod_i Q_i^{\beta_i}$. Write $Z = 1 + W$.

Definition 8.1. The *free energy* is $F = \log Z = \sum_{m \geq 1} \frac{(-1)^{m+1}}{m} W^m = \sum_{\beta \neq 0} F_\beta Q^\beta$.

Since W has no constant term, only finitely many m contribute to each F_β (at most the total degree of β). For one variable:

$$F_1 = Z_1, \qquad F_2 = Z_2 - \tfrac{1}{2} Z_1^2, \qquad F_3 = Z_3 - Z_1 Z_2 + \tfrac{1}{3} Z_1^3. \tag{55}$$

For two variables one has, for example, $F_{(1,1)} = Z_{(1,1)} - Z_{(1,0)} Z_{(0,1)}$ and $F_{(2,0)} = Z_{(2,0)} - \frac{1}{2} Z_{(1,0)}^2$.

8.2 The Gopakumar-Vafa expansion

Definition 8.2 (GV invariants, physical convention)**.** The *Gopakumar-Vafa invariants* n_β^g ($g \geq 0$, $\beta \neq 0$) are the numbers defined by

$$F = \sum_{\beta \neq 0} \sum_{g \geq 0} \sum_{k \geq 1} n_\beta^g \, (-1)^{g-1} \, \frac{1}{k} \, [k]^{2g-2} \, Q^{k\beta}. \tag{56}$$

The existence of such numbers, their integrality, and the fact that for each β only finitely many g have $n_\beta^g \neq 0$ is a theorem. It was proved in exactly the combinatorial setting of these notes, by Peng [3] for local surfaces and by Konishi [2] in general. The algorithm below computes the n_β^g and, along the way, checks these properties.

Physics remark (motivation only, may be skipped). In physics $q = e^{ig_s}$ with g_s the string coupling, and $F = \sum_g g_s^{2g-2} F_g(Q)$ is the genus expansion of the topological string free energy. Gopakumar and Vafa [9] argued that F has the form $\sum n_\beta^g \frac{1}{k}\big(2\sin\frac{kg_s}{2}\big)^{2g-2} Q^{k\beta}$. Here n_β^g counts (with signs) BPS states of M2-branes wrapping curves of class β and "genus" g, and the sum over k accounts for k-fold multiple covers. With $q = e^{ig_s}$ we have $[k] = q^{k/2} - q^{-k/2} = 2i\sin\frac{kg_s}{2}$, so $\big(2\sin\frac{kg_s}{2}\big)^2 = -[k]^2$ and $\big(2\sin\frac{kg_s}{2}\big)^{2g-2} = (-1)^{g-1}[k]^{2g-2}$, which is (56). The leading power $[k]^{-2} \sim (kg_s)^{-2}$ for $g = 0$ is the genus-zero contribution, and higher g come with more powers of g_s. The variable $y = [1]^2 = -4\sin^2\frac{g_s}{2}$ is the natural expansion parameter.

8.3 Extracting the integers

Fix β and suppose all $n_{\beta'}^g$ for classes $\beta' = \beta/k$ ($k \geq 2$, $k \mid \beta$, meaning k divides every component of β) are already known. In (56) the coefficient of Q^β receives contributions from $(\beta, k = 1)$ and from $(\beta/k, k)$. Define

$$f_\beta = F_\beta - \sum_{k\geq 2,\ k\mid\beta} \frac{1}{k} \sum_g n_{\beta/k}^g (-1)^{g-1}[k]^{2g-2}. \tag{57}$$

Then $f_\beta = \sum_g n_\beta^g (-1)^{g-1}[1]^{2g-2}$, i.e.

$$P_\beta := y\, f_\beta = \sum_{g\geq 0} n_\beta^g (-1)^{g-1} y^g, \qquad y = [1]^2 = q - 2 + q^{-1}. \tag{58}$$

Algorithm 2 (GV extraction)**.**
Input: F_β for all β up to some degree. **Output:** n_β^g.
Process the classes β in order of increasing degree. For each β:

1. Compute f_β by (57) (the subtracted terms use invariants of smaller classes).
2. Compute $P_\beta = y\, f_\beta$ and simplify. *Check:* P_β is a Laurent polynomial in q (no denominators, integral powers of q) and $P_\beta(q) = P_\beta(1/q)$.
3. Write P_β as a polynomial in y by peeling off the top power: if the highest power of q in P_β is q^G with coefficient c_G, subtract $c_G y^G$ (the top term of y^G is q^G) and repeat (Exercise 3.3).
4. If $P_\beta = \sum_g c_g y^g$, then $n_\beta^g = (-1)^{g-1} c_g$. *Check:* integers.

8.4 Examples

Conifold. By Proposition 7.2, $F_m = -\frac{1}{m[m]^2}$. For $\beta = 1$: $P_1 = yF_1 = -1 = n_1^0(-1)^{-1}$, so $n_1^0 = 1$. For $\beta = m \geq 2$: $f_m = F_m - \frac{1}{m} n_1^0 (-1)^{-1}[m]^{-2} = 0$, so $n_m^g = 0$. The conifold has a single BPS state: one rigid sphere.

Local $\mathbb{P}^2$, degree 1. $F_1 = Z_1 = -3/y$, so $P_1 = -3$ and $n_1^0 = 3$: the three "lines" e_1, e_2, e_3 each contribute one. (Geometrically, the lines in $\mathbb{P}^2$ form a 2-dimensional family, and $n_1^0 = 3$ is its signed Euler characteristic.)

Local $\mathbb{P}^2$, degree 2. From $Z_1 = -3/y$ and (51),

$$F_2 = Z_2 - \tfrac{1}{2}Z_1^2 = \frac{6y^2+27y+18}{y^2(y+4)} - \frac{9}{2y^2} = \frac{12y^2+54y+36-9y-36}{2y^2(y+4)} = \frac{12y+45}{2y(y+4)}. \tag{59}$$

The class $2 = 2\cdot 1$ is divisible by $k=2$, and $n_1^0 = 3$, so the multicover subtraction is

$$f_2 = F_2 - \frac{1}{2}\,n_1^0\,(-1)^{-1}[2]^{-2} = F_2 + \frac{3}{2y(y+4)} = \frac{12y+48}{2y(y+4)} = \frac{6}{y}. \tag{60}$$

Therefore $P_2 = yf_2 = 6 = n_2^0(-1)^{-1}$, i.e. $n_2^0 = -6$, and $n_2^g = 0$ for $g \geq 1$. Note how the denominator $y+4$ (coming from $[2]^2$) cancels only after the subtraction: this is the integrality theorem at work.

Local $\mathbb{P}^2$, degree 3. Here $k=3$ divides 3, and $f_3 = F_3 - \frac{1}{3}n_1^0(-1)^{-1}[3]^{-2} = F_3 + \frac{1}{[3]^2}$. With F_3 from the generated block in Section 7.2 one finds $P_3 = yf_3 = -10y-27$, so $n_3^0 = (-1)^1\cdot(-27) = 27$ and $n_3^1 = (-1)^0\cdot(-10) = -10$. This is the first genus-one invariant: plane cubics are elliptic curves.

Local $\mathbb{P}^1\times\mathbb{P}^1$. From Section 7.3: $F_{(1,0)} = 2/y$, so $n^0_{(1,0)} = -2$ (a $\mathbb{P}^1$-family of $(0,-2)$-curves: signed Euler characteristic -2). Next $F_{(2,0)} = Z_{(2,0)} - \frac{1}{2}Z^2_{(1,0)} = \frac{3y+8}{y^2(y+4)} - \frac{2}{y^2} = \frac{1}{y(y+4)} = \frac{1}{[2]^2}$, and $f_{(2,0)} = F_{(2,0)} - \frac{1}{2}(-2)(-1)[2]^{-2} = 0$, so $n^g_{(2,0)} = 0$. Finally $F_{(1,1)} = Z_{(1,1)} - Z_{(1,0)}Z_{(0,1)} = \frac{4(y+1)}{y^2} - \frac{4}{y^2} = \frac{4}{y}$ (no multicover term, since $(1,1)$ is not divisible by any $k\geq 2$), so $n^0_{(1,1)} = -4$.

8.5 Tables

Table 3 lists the invariants of local $\mathbb{P}^2$ up to degree 5 and the genus-zero invariants of local $\mathbb{P}^1\times\mathbb{P}^1$ up to bidegree $(3,3)$, as given in the literature (KKV [8], Table 4, and CKYZ [7], Table 9, in the physical convention). The package reproduces all of them from Algorithm 1 and Algorithm 2 (its test suite compares with these numbers, stored in `tests/data.py`).

d	$g=0$	$g=1$	$g=2$	$g=3$	$g=4$	$g=5$	$g=6$
1	3	0	0	0	0	0	0
2	−6	0	0	0	0	0	0
3	27	−10	0	0	0	0	0
4	−192	231	−102	15	0	0	0
5	1695	−4452	5430	−3672	1386	−270	21

$d_B\backslash d_F$	0	1	2	3
0	–	−2	0	0
1	−2	−4	−6	−8
2	0	−6	−32	−110
3	0	−8	−110	−756

Table 3: (Generated from `tests/data.py`.) Left: n_d^g for local $\mathbb{P}^2$, $d\leq 5$, all genera (all n_d^g not shown vanish, and the top genus $g=(d-1)(d-2)/2$ has $n_d^g = (-1)^{d(d+3)/2}\big(\frac{d(d+3)}{2}+1\big)$, e.g. $n_5^6 = 21$). Right: $n^0_{(d_B,d_F)}$ for local $\mathbb{P}^1\times\mathbb{P}^1$, symmetric under $d_B\leftrightarrow d_F$.

9 Curve classes and truncation

In this section we explain which of the variables Q_e have to be identified, and how the sum (39) is truncated.

9.1 The class of an edge

Let $\omega_1, \dots, \omega_N$ be all the lattice points of the polygon (in some fixed order) and $\widehat{\omega}_i = (a_i, b_i, 1)$.

Definition 9.1. The *relation lattice* is $L = \{l \in \mathbb{Z}^N : \sum_i l_i \widehat{\omega}_i = 0\}$. For an interior segment τ, with $\omega_1, \omega_2, \omega_3, \omega_3'$ and a_1, a_2 as in Lemma 6.1, the *relation vector* (or *class*) $l_\tau \in L$ has entries a_1 at ω_1, a_2 at ω_2, $+1$ at ω_3 and at ω_3', and 0 elsewhere. The statement $l_\tau \in L$ is precisely the relation (35).

The vector l_τ does not depend on which of the two flags is used (the two choices exchange $\omega_1 \leftrightarrow \omega_2$, $a_1 \leftrightarrow a_2$ and $\omega_3 \leftrightarrow \omega_3'$). Since the $\widehat{\omega}_i$ span a space of dimension 3, L has rank $N-3$.

Physics remark (motivation only, may be skipped). $L \cong H_2(X;\mathbb{Z})$, and l_τ is the homology class of the curve C_τ: its entries are the intersection numbers of C_τ with the toric divisors D_i (one per lattice point). For the interior point ω_0 of a star triangulation, D_0 is the compact surface S and the entry of l_τ at ω_0 is $-$(self-intersection of C_τ in S) $-\,2$, consistent with Remark 7.3 (e.g. -3 for a line in $\mathbb{P}^2$).

The term of (39) with partitions (λ_τ) describes curves in the class $\beta = \sum_\tau |\lambda_\tau|\, l_\tau \in L$. Therefore, after choosing a basis $b_1, \dots, b_r$ of L ($r = N-3$) and writing $l_\tau = \sum_i c_{\tau i} b_i$, we substitute

$$Q_\tau \;\longmapsto\; \prod_{i=1}^{r} Q_i^{c_{\tau i}}. \tag{61}$$

Since (61) is a ring homomorphism it commutes with log, so it may be done before or after taking $F = \log Z$ (in [2, (8)] the logarithm is taken first, and then one sums over the degree vectors in a given class).

Example 9.2.

- *Local* $\mathbb{P}^2$ (points $\omega_0, \dots, \omega_3$ as in Section 7.2). For $\tau = \omega_0\omega_1$ we found $\omega_3' = -a_1\omega_0 - a_2\omega_1 - \omega_2$ with $a_1 = -3$ (at ω_0) and $a_2 = 1$ (at ω_1), so $l = (-3, 1, 1, 1)$. One can check that $-3(0,0,1) + (1,0,1) + (0,1,1) + (-1,-1,1) = 0$. By symmetry all three segments have the same class H (the line in $\mathbb{P}^2$), so $Q_1 = Q_2 = Q_3 = Q$.
- *Local* $\mathbb{P}^1 \times \mathbb{P}^1$. Here $l(\omega_0\omega_1) = l(\omega_0\omega_3) = (-2, 0, 1, 0, 1)$ and $l(\omega_0\omega_2) = l(\omega_0\omega_4) = (-2, 1, 0, 1, 0)$. The rank is $5 - 3 = 2$, with two classes B and F, and $Q_1 = Q_3 = Q_B$, $Q_2 = Q_4 = Q_F$.
- *Local* F_1 (points $\omega_0, \dots, \omega_4 = (0,0), (0,1), (-1,1), (0,-1), (1,0)$):

$$l_1 = (-1,-1,1,0,1), \quad l_2 = l_4 = (-2,1,0,1,0), \quad l_3 = (-3,0,1,1,1) = l_1 + l_2. \tag{62}$$

  Therefore, with $B = l_1$ (the (-1)-curve) and $F = l_2$ (the fiber), the four edges have classes $B, F, B+F, F$, and $Q_1 = Q_B$, $Q_2 = Q_4 = Q_F$, $Q_3 = Q_B Q_F$.

9.2 Truncation needs positive weights

To compute F_β we need *all* assignments (λ_τ) with $\sum_\tau |\lambda_\tau| l_\tau = \beta$, and we need the products in $\log Z$ to be complete. The naive cutoff "total edge degree $\sum_\tau |\lambda_\tau| \le D$" is not adapted to classes:

Example 9.3 (Local F_1)**.** The class $B+F$ arises from $\lambda_3=(1)$ (edge degree 1) and from $\lambda_1=\lambda_2=(1)$ or $\lambda_1=\lambda_4=(1)$ (edge degree 2). With the cutoff "edge degree ≤ 1" one would get $F_{B+F}=Z_{B+F}=-1/y$ and conclude $n^0_{B+F}=1$. The correct value, using all terms (Exercise 7.2), is $F_{B+F}=-3/y$ and $n^0_{B+F}=3$.

The cure is to give each edge a positive *weight* w_τ that is a linear function of its class, and truncate by $\sum_\tau w_\tau|\lambda_\tau|\le D$. Then the weight $w(\beta)$ of a class is well defined, the truncation keeps exactly the classes with $w(\beta)\le D$, and keeps them completely. Moreover, since all $w_\tau>0$, each class receives only finitely many terms.

Such weights are provided by heights of the triangulation (Definition 2.14). Namely, for heights $h_1,\dots,h_N$ take

$$w_\tau := \langle h, l_\tau\rangle = \sum_i h_i\,(l_\tau)_i. \tag{63}$$

This is the weight of τ introduced in Section 2.5. To see this, let ϕ be the affine function on σ with $\phi(\omega_i)=h_i$ at its corners. Then $\sum_i(l_\tau)_i\phi(\omega_i)=0$ because $l_\tau\in L$, so that $\langle h,l_\tau\rangle=h(\omega_3')-\phi(\omega_3')$. This is the height of the lifted fourth point above the plane through the lifted triangle σ. In particular, the integers $h_1,\dots,h_N$ are heights of the triangulation if and only if $w_\tau>0$ for all interior segments τ. The weights are linear in the class by construction. With the heights $h=0$ at the interior point and $h=1$ at the boundary points (Example 2.15), which can also be used for local F_1, one gets

$$\mathbb{P}^2:\ w=3,\qquad \mathbb{P}^1\times\mathbb{P}^1:\ w_B=w_F=2,\qquad F_1:\ w_B=1,\ w_F=2,\ w_{B+F}=3. \tag{64}$$

For F_1 the truncation $\sum w_\tau|\lambda_\tau|\le 3$ contains the class $B+F$ completely. The package uses the weights (63) by default, with the heights found as described in Section 2.5.

Physics remark (motivation only, may be skipped). w_τ is the area of the curve C_τ for the Kähler class determined by h, and regularity of the triangulation is the statement that X admits a Kähler metric (it is a projective morphism over the singular affine toric variety). Truncating by area is natural: $Q^\beta=e^{-t(\beta)}$.

A Cross-check: the 5d Nekrasov partition function

In this appendix we compare the vertex partition function with a completely different formula: the Nekrasov instanton partition function of a 5d gauge theory [12, 6]. The comparison is possible for a large class of triangulations. The gauge theories that appear are *linear quivers*: the gauge group is $U(N_1)\times\ldots\times U(N_r)$, with bifundamental matter between consecutive factors, fundamental matter at the two ends, and Chern-Simons terms. The dictionary between the triangulation and the gauge theory given below was read off from the dual graph and verified by computer (`topvertex.nekrasov`). It is not taken from the literature.

Ladder triangulations. Call a triangulation a *ladder* along a primitive vector d if every triangle has a side parallel to d. After an orientation-preserving unimodular change of coordinates we may take $d=(0,1)$. The lattice points of the polygon are then arranged in columns $X=0,1,\dots,r+1$,

$$\text{column } i:\qquad \omega^{(i)}_j=(i,\,b_i+j),\qquad j=0,\dots,N_i. \tag{65}$$

The triangulation contains all the vertical unit segments $\omega^{(i)}_{k-1}\omega^{(i)}_k$, and every strip between two adjacent columns is cut into triangles that have one vertical side and the opposite corner in the other column (Figure 2). We assume $r\ge 1$.

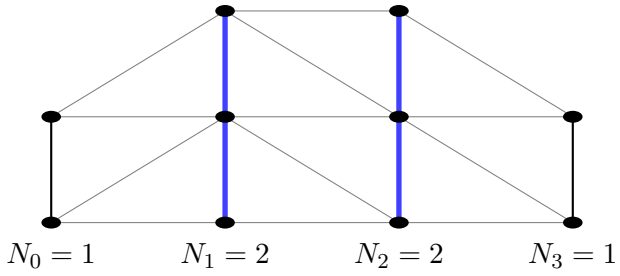


Figure 2: A ladder triangulation with $r = 2$ and all $b_i = 0$: the gauge theory is $U(2) \times U(2)$ with one bifundamental and one fundamental hypermultiplet at each end, and $\kappa_1 = \kappa_2 = -\frac{1}{2}$. The four thick blue segments are the gauge branes. They are dual to the horizontal edges that carry the partitions $Y_k^{(i)}$.

The gauge theory. The columns $1, \dots, r$ are the gauge groups $U(N_i)$. Two adjacent gauge columns are joined by a bifundamental hypermultiplet. The columns 0 and $r+1$ give N_0 and N_{r+1} fundamental hypermultiplets of the first and of the last group (none if the column is a single point). In the dual graph the vertical segment $\omega_{k-1}^{(i)}\omega_k^{(i)}$ is a horizontal edge, which we call the *brane* (i,k), $k = 1, \dots, N_i$ counted from the bottom. For $1 \le i \le r$ it is an internal edge, of class $T_{i,k}$, and it carries a partition $Y_k^{(i)}$. For $i = 0, r+1$ it is an external leg and we put $Y_k^{(i)} = \varnothing$.

The position of the brane (i,k) is the vector

$$y_k^{(i)} = e_{\omega_k^{(i)}} - e_{\omega_{k-1}^{(i)}} \in \mathbb{Z}^{\#\text{points}}, \qquad \text{so that} \qquad \langle h, y_k^{(i)} \rangle = h(\omega_k^{(i)}) - h(\omega_{k-1}^{(i)}) \tag{66}$$

for heights h of the triangulation (Section 2.5). Two facts make this useful. First, the difference $y - y'$ of two branes in the same column or in adjacent columns is a relation vector, and therefore a curve class. We write $Q^{y-y'}$ for the corresponding monomial. Second, its Kähler weight $\langle h, y - y' \rangle$ is never zero, and its sign does not depend on the choice of h. Namely, the triangles of a strip are ordered from bottom to top, and each of them contains exactly one brane. Two consecutive triangles are separated by a diagonal segment τ, and

$$y(\text{upper brane}) - y(\text{lower brane}) = l_\tau. \tag{67}$$

Therefore the triangulation of a strip is the same thing as the vertical order of its branes, that is a *chamber* of the gauge theory (different ladder triangulations of the same polygon are related by flops). The monomial $Q^{y-y'}$ is the product of the Q_τ over the diagonals between the two branes. We write $|D|$ for the one of $\pm D$ that has positive weight.

The formula. For partitions Y, Y' and a box $s = (i,j)$ put $a_Y(s) = Y_i - j$ and $l_{Y'}(s) = Y'^{\mathsf{t}}_j - i$ (arm and leg measured in the indicated partition, and they may be negative when s lies outside it). Put $a_k^{(i)} = -y_k^{(i)}$ and, for two branes in the same or in adjacent columns, with $D = y' - y$, define the "shifted q-number"

$$[a - a' + c] := Q^{-D/2}x^c - Q^{D/2}x^{-c} = \pm\, Q^{-|D|/2}\, x^{\pm c}\left(1 - Q^{|D|}q^{\mp c}\right) \tag{68}$$

(upper signs if D has positive weight), to be expanded in powers of $Q^{|D|}$. For $a = a'$ it is the ordinary $[c]$. For two partitions let

$$n_{\mu\nu}(u) = \prod_{s\in\mu} \big[u + a_\mu(s) + l_\nu(s) + 1\big] \prod_{s\in\nu} \big[u - a_\nu(s) - l_\mu(s) - 1\big]. \tag{69}$$

The partition function of the quiver is

$$Z^{\text{quiver}} = \sum_{Y} \prod_{i=1}^{r} \frac{\phi_i^{|Y^{(i)}|}\ \mathrm{CS}_{\kappa_i}(Y^{(i)})}{\prod_{k,l=1}^{N_i} n_{Y_k^{(i)} Y_l^{(i)}}\big(a_k^{(i)} - a_l^{(i)}\big)} \ \prod_{i=0}^{r} \prod_{k=1}^{N_i} \prod_{l=1}^{N_{i+1}} n_{Y_k^{(i)} Y_l^{(i+1)}}\big(a_k^{(i)} - a_l^{(i+1)}\big), \tag{70}$$

where $|Y^{(i)}| = \sum_k |Y_k^{(i)}|$ and $\mathrm{CS}_\kappa(Y^{(i)}) = \prod_k \prod_{(a,b)\in Y_k^{(i)}} \big(Q^{\,y_1^{(i)} - y_k^{(i)}}\, q^{a-b}\big)^{\kappa}$ (the frame $a_1^{(i)} = 0$ at every node). The first product contains the vector multiplets, and the second one the bifundamental ($1 \le i < r$) and fundamental ($i = 0, r$) hypermultiplets. The vector multiplet factor is the arm/leg form of the i,j-product in the formula (74) of [6] (checked by the package).

The dictionary. The Chern-Simons levels, the signs and the instanton parameters are

$$\kappa_i = b_{i+1} - 2b_i + b_{i-1} - N_i + \tfrac{1}{2}(N_{i+1} + N_{i-1}), \qquad \varepsilon_i = (-1)^{b_{i+1}+b_{i-1}+N_i+N_{i-1}}, \tag{71}$$

$$\phi_i = \varepsilon_i\ Q_{T_{i,1}} \prod_{k=2}^{N_i} Q^{-(y_k^{(i)} - y_1^{(i)})} \prod_{l=1}^{N_{i-1}} Q^{\frac{1}{2}|y_1^{(i)} - y_l^{(i-1)}|} \prod_{l=1}^{N_{i+1}} Q^{\frac{1}{2}|y_1^{(i)} - y_l^{(i+1)}|}. \tag{72}$$

With these values the leading monomial of the term Y of (70) is exactly $\prod_{i,k} Q_{T_{i,k}}^{|Y_k^{(i)}|}$, with integer exponents: the partitions of the Nekrasov sum are the partitions on the horizontal edges of the vertex sum, and all the other edges have been summed. To compare, define the *instanton numbers* of a class with relation vector l by $n_i(l) = \sum_\omega l_\omega \max(0, X_\omega - i)$, $i = 1, \dots, r$. They satisfy $n_j(T_{i,k}) = \delta_{ij}$ and $n_j = 0$ on the diagonal edges. Let $Z^{\text{vertex}}_{\text{pert}}$ be the part of Z^{vertex} with all $n_i = 0$. Then

$$\frac{Z^{\text{vertex}}}{Z^{\text{vertex}}_{\text{pert}}} = Z^{\text{quiver}}. \tag{73}$$

Remark A.1. (i) κ_i is a half-integer exactly when $N_{i-1} + N_{i+1}$ is odd (the parity anomaly of $U(N_i)$ with an odd number of flavors), and then ϕ_i contains square roots of the Q's, which cancel in every term. (ii) The formula for ε_i is not symmetric under $i - 1 \leftrightarrow i + 1$ because (70) orients the bifundamentals from left to right: $n_{\nu\mu}(-u) = (-1)^{|\mu|+|\nu|} n_{\mu\nu}(u)$. (iii) Rotating the picture by π (the ladder along $-d$) reverses the order of the columns and of the branes and replaces κ_i by $-\kappa_i$. Both descriptions are valid.

Example: pure $SU(N)$. The polygon (`topvertex.examples_geometry.su_n_ladder(N, m)`)

$$P_{N,m} = \mathrm{conv}\big(\{c_j = (0,j)\}_{0\le j\le N} \cup \{L = (-1,1),\ R = (1, N-m-1)\}\big), \qquad 0 \le m \le N, \tag{74}$$

triangulated by $\{c_j, c_{j+1}, L\}$ and $\{c_j, c_{j+1}, R\}$, $0 \le j < N$, is a ladder along $(0,1)$ with the three columns $\{L\}$, $\{c_0, \dots, c_N\}$, $\{R\}$. Therefore $r = 1$, $N_0 = N_2 = 0$, $N_1 = N$ and $b = (1, 0, N-m-1)$: there is no matter, and

$$\kappa = -m, \qquad \varepsilon = (-1)^m. \tag{75}$$

For $N = 2$ and $m = 0, 1, 2$ the polygon is local $\mathbb{P}^1 \times \mathbb{P}^1$, F_1, F_2. The triangles of the left strip are separated by the diagonals $c_j L$, $1 \le j < N$. Let F_j be the class of $c_j L$ (the "W-boson" curves). Then $y_n - y_l = F_l + \ldots + F_{n-1}$ for $l < n$, so with $Q_{ln} = \prod_{k=l}^{n-1} Q_{F_k}$ the shifted q-numbers are $[a_l - a_n + c] = Q_{ln}^{-1/2} x^c - Q_{ln}^{1/2} x^{-c}$, and (70) becomes

$$Z^{\text{quiver}} = \sum_{Y_1,\dots,Y_N} \frac{\phi^{\sum_l |Y_l|}\ \mathrm{CS}_\kappa(Y)}{\prod_{l,n=1}^{N} n_{Y_l Y_n}(a_l - a_n)}, \qquad \mathrm{CS}_\kappa(Y) = \prod_{l=1}^{N} \prod_{(a,b)\in Y_l} \Big(q^{a-b} \prod_{j<l} Q_{F_j}^{-1}\Big)^{\kappa}, \tag{76}$$

which is Nekrasov's partition function of pure $SU(N)$ gauge theory with Chern-Simons level κ [12, 6], written in the frame $a_1 = 0$. The gauge branes are the column segments $c_{k-1}c_k$. Let $B_{\rm bot} = T_{1,1}$ be the class of the bottom one. There are no flavor branes, so (72) reads

$$\phi = (-1)^m\, Q_{B_{\rm bot}} \prod_{j=1}^{N-1} Q_{F_j}^{-(N-j)}, \tag{77}$$

and $Z^{\rm vertex}_{\rm pert}$ is the contribution of the fiber curves alone. In the frame $\sum_l a_l = 0$ the exponents become $-(N-m)(N-j)/N$, so for $m = N$ one has $\phi = \pm Q_{B_{\rm bot}}$ exactly.

One instanton for $N = 2$. For $N = 2$, (76) at order ϕ (the two terms $Y = ((1), \varnothing)$ and $(\varnothing, (1))$) is

$$Z_1^{\rm Nek} = \frac{(1 + Q_F^{-\kappa})\, Q_F}{[1]^2 (1 - Q_F)^2} = \begin{cases} \sum_{d\geq 1} 2d\, Q_F^d/[1]^2, & \kappa = 0, \\ \sum_{d\geq 1} (2d-1)\, Q_F^d/[1]^2, & \kappa = -1. \end{cases} \tag{78}$$

Multiplying by $\phi = (-1)^m Q_B Q_F^{-1}$ gives the order-Q_B part of $F = \log Z^{\rm vertex}$: $\sum_{d\geq 0} 2(d+1) Q_B Q_F^d/[1]^2$ for local $\mathbb{P}^1 \times \mathbb{P}^1$ ($m = 0$) and $-\sum_{d\geq 0}(2d+1) Q_B Q_F^d/[1]^2$ for local F_1 ($m = 1$, B the (-1)-curve). By (56) these are $n^0_{B+dF} = -2(d+1)$ for local $\mathbb{P}^1 \times \mathbb{P}^1$, in agreement with the row $d_B = 1$ of Table 3, and $n^0_{B+dF} = 2d+1$ for local F_1, in agreement with Exercise 7.2 ($n^0_B = 1$, $n^0_{B+F} = 3$). (Here $B = B_{\rm bot}$, which for $m = 1$ is the segment with framing 0, i.e. the (-1)-curve.) In the $SU(2)$ frame these identifications read $\phi = Q_B/Q_F$ ($m = 0$) and $\phi = -Q_B Q_F^{-1/2}$ ($m = 1$).

Example: $U(1)$. The smallest ladders have one gauge brane. The triangle with corners $(0,0)$, $(2,0)$, $(1,1)$ and the segment from $(1,0)$ to $(1,1)$ has $b = (0,0,0)$ and $N = (0,1,0)$, so $\kappa = -1$, $\varepsilon = -1$ and $\phi = -Q$. Here $h(s)$ is the hook length, $n_{YY}(0) = (-1)^{|Y|}\prod_{s\in Y}[h(s)]^2$ and $\mathrm{CS}_{-1}(Y) = q^{-\sum_{(a,b)\in Y}(a-b)} = q^{\kappa(Y)/2}$, so that (70) and the hook formula (20) give

$$Z^{\rm quiver} = \sum_Y Q^{|Y|}\, \frac{q^{\kappa(Y)/2}}{\prod_{s\in Y}[h(s)]^2} = \sum_Y Q^{|Y|} s_Y(q^\rho)^2. \tag{79}$$

This is the one-edge partition function with framing $n = 1$ of Section 7.1 ($\mathcal{O}(0)\oplus\mathcal{O}(-2) \to \mathbb{P}^1$). Moving the left corner to $(0,1)$ gives the conifold: $b = (1,0,0)$, $\kappa = 0$, $\varepsilon = +1$, and $Z^{\rm quiver} = \sum_Y (-Q)^{|Y|}\prod_{s\in Y}[h(s)]^{-2}$ is the one-edge partition function with $n = 0$. In both cases $Z^{\rm vertex}_{\rm pert} = 1$.

Checks. Table 4 lists some of the geometries for which $Z^{\rm vertex} = Z^{\rm vertex}_{\rm pert}\, Z^{\rm quiver}$ was verified exactly, coefficient by coefficient, for all classes of Kähler weight ≤ 6 (in three chambers each when the polygon has several ladder triangulations). From the command line, the option `--nekrasov` runs this check up to the chosen `--order`. In Python the same check is done by `topvertex.nekrasov.compare_with_nekrasov(triangulation, max_weight)`.

B Conventions dictionary

In this appendix we relate our conventions, which are the conventions of [2], to the ones used in [1, 4, 5].

columns (b_i, N_i), $i = 0, \dots, r+1$	gauge theory	κ_i	ε_i
$(1,0),(0,N),(N-m-1,0)$	$U(N)$, $N \le 4$, $0 \le m \le N$	$-m$	$(-1)^m$
$(1,0),(0,2),(2,0)$	$U(2)$	1	$-$
$(3,0),(0,3),(3,0)$	$U(3)$	3	$-$
$(0,1),(0,2),(1,0)$	$U(2)$, $N_f = 1$	$-\frac{1}{2}$	$+$
$(0,2),(0,2),(0,2)$	$U(2)$, $N_f = 2+2$	0	$+$
$(1,1),(0,3),(1,1)$	$U(3)$, $N_f = 1+1$	0	$+$
$(0,0),(0,1),(0,1),(0,1),(0,0)$	$U(1)^3$	$-\frac{1}{2}, 0, -\frac{1}{2}$	$-,+,+$
$(1,0),(0,2),(0,3),(2,0)$	$U(2) \times U(3)$	$\frac{1}{2}, 0$	$-,-$
$(0,1),(0,2),(0,2),(0,1)$	$U(2) \times U(2)$, $N_f = 1+1$	$-\frac{1}{2}, -\frac{1}{2}$	$-,+$
$(0,3),(0,2),(0,1),(0,0)$	$U(2) \times U(1)$, $N_f = 3+0$	$0, 0$	$-,-$
$(1,0),(0,2),(0,2),(0,2),(1,0)$	$U(2)^3$	$0,0,0$	$-,+,-$
$(2,0),(1,2),(0,3),(1,2),(2,0)$	$U(2) \times U(3) \times U(2)$	$-\frac{1}{2}, 1, -\frac{1}{2}$	$+,-,-$

Table 4: Linear quivers checked against the vertex. The polygon is the convex hull of the points $(i, b_i + j)$, $0 \le j \le N_i$. The first row is $P_{N,m}$, and the row $U(2) \times U(1)$ is the triangle with corners $(0,0),(3,0),(0,3)$.

The variable q. We follow [2]: $q = e^{ig_s}$ ("physical" convention), so that $[k] = 2i\sin(kg_s/2)$ and $\big(2\sin\frac{kg_s}{2}\big)^{2g-2} = (-1)^{g-1}[k]^{2g-2}$, which is how (56) arises. AKMV [1] and Mariño [4] write $q = e^{g_s}$, so that $[k] = 2\sinh(kg_s/2)$, and expand $F = \sum \tilde{n}^g_\beta \frac{1}{k}\big(2\sinh\frac{kg_s}{2}\big)^{2g-2} Q^{k\beta}$. As formal rational functions of $q^{1/2}$ the partition functions are the same. Comparing with the expansion (56) gives $\tilde{n}^g_\beta = (-1)^{g-1} n^g_\beta$. (Tables in the literature may contain further, class-dependent signs, e.g. from a different sign convention for the variables Q. The tests of the package determine them case by case.)

The vertex. Definition 5.1 is [2, Def. 2.2]. In [5, (7)] the vertex is written as

$$C_{\lambda\mu\nu} = q^{\kappa(\lambda)/2} s_\nu(q^\rho) \sum_\eta s_{\lambda^{\mathrm{t}}/\eta}(q^{\nu+\rho})\, s_{\mu/\eta}(q^{\nu^{\mathrm{t}}+\rho}). \tag{80}$$

Substituting $(\lambda_1,\lambda_2,\lambda_3) = (\mu,\nu,\lambda)$ in (26) gives exactly this expression, so their $C_{\lambda\mu\nu}$ is our $C_{\mu\nu\lambda}$, which equals our $C_{\lambda\mu\nu}$ by cyclic symmetry: it is the same function. In [1] the vertex is defined through Chern-Simons theory. The closed formula in terms of Schur functions is due to [10]. All values with one or two boxes listed in [1, §8] coincide with Table 2 (and with those quoted in [2, p. 6]). Of the three-box entries, four coincide exactly with ours:

$$\frac{q^{9/2}}{(q-1)(q^2-1)(q^3-1)} = C_{(3)\varnothing\varnothing}, \qquad \frac{q^{5/2}}{(q-1)^2(q^3-1)} = C_{(2,1)\varnothing\varnothing}, \tag{81}$$

$$\frac{q^{3/2}}{(q-1)(q^2-1)(q^3-1)} = C_{(1,1,1)\varnothing\varnothing}, \qquad \frac{q^{3/2}(q^3-q^2+1)}{(q-1)^2(q^2-1)} = C_{(2)(1)\varnothing}. \tag{82}$$

The fifth printed three-box value, $\frac{q^{-1/2}(q^3-q^2+1)}{(q-1)^2(q^2-1)}$, equals $q^{-2}C_{(2)(1)\varnothing} = q^{-\kappa((2))}C_{(2)(1)\varnothing}$: it differs from ours by a framing factor, consistent with AKMV defining the vertex through $W_{R_1R_2}$ in a different framing/orientation convention. (The diagram labels of their table are not legible in the

text version we used, so we cannot say which slot assignment that entry corresponds to.) We do not use their tables.

The gluing rule. Our rule is the following. The edge e goes from the tail a to the head b, the label n_e is the framing at the tail, and λ_e is untransposed at a and transposed at b. The edge factor is $(-1)^{|\lambda_e|(n_e+1)}q^{+n_e\kappa(\lambda_e)/2}$, and the triples are read counterclockwise. The formula (3.15) of [1] reads

$$\sum_R C_{R_jR_kR}\, e^{-|R|t}\,(-1)^{(n+1)|R|}\,q^{-n\kappa(R)/2}\,C_{R^{\mathsf{t}}R'_jR'_k}, \qquad n=|v_{k'}\wedge v_k|, \tag{83}$$

where the edge (leg i) is outgoing at the first vertex, (v_i,v_j,v_k) and $(v_i,v_{j'},v_{k'})$ are the counterclockwise triples at the two vertices, and $|v_{k'}\wedge v_k|$ is $v_{k'}\wedge v_k$ if $v_k, v_{k'}$ are both outgoing (or both incoming) and $-v_{k'}\wedge v_k$ otherwise. Structurally this is our rule (untransposed at the vertex where the edge is outgoing, transposed at the other end, the same sign). The sign of the exponent requires care: with the standard determinant $(p_1,q_1)\wedge(p_2,q_2)=p_1q_2-q_1p_2$ and outgoing leg vectors, $v_{k'}\wedge v_k$ is *our* n_e (Remark 6.10), and our factor is $q^{+n_e\kappa/2}$. The two agree if AKMV's $\wedge$ is the opposite of the standard determinant, which is what their normalization $v_2\wedge v_1=v_1\wedge v_3=v_3\wedge v_2=1$ for counterclockwise legs [1, §3.4] suggests (the standard determinant gives $v_1\wedge v_2=v_2\wedge v_3=v_3\wedge v_1=+1$ for counterclockwise legs). The paper [5] uses $v_i\wedge v_{i+1}=1$ and also writes $q^{-n\kappa/2}$ with $n=v'_3\wedge v_3$ in its formula (10). The sign matters: replacing every n_e by $-n_e$ in our rule changes the result (for local $\mathbb{P}^2$ it would give $n_3^0=15$, $n_3^1=-4$ instead of $27,-10$). We have therefore fixed all signs by the internal consistency of [2] and by the comparison with [8, 7], and recommend that anyone importing a formula from the literature test it on local $\mathbb{P}^2$.

D-branes. In [1] the vertex is the amplitude of three stacks of *D-branes* (boundary conditions for open strings) on the three legs of $\mathbb{C}^3$, and gluing "annihilates" a brane with an anti-brane, which is the origin of the sign $(-1)^{|R|}$ in (3.14) of [1]. None of this is needed for the closed-string partition function computed here.

C Answers to the exercises

In this appendix we collect the answers to the exercises.

Exercise 3.1. $(3,2)^{\mathsf{t}}=(2,2,1)$. The contents are $0,1,2\,|\,-1,0$, so $\kappa=2\cdot 2=4$, and also $3\cdot 2+2\cdot(2-4+1)=4$. The hook lengths are $4,3,1\,|\,2,1$, and $120/24=5$.

Exercise 3.2. $\sum_i(2i-1)\lambda_i=\sum_{(i,j)\in\lambda}(2i-1)=\sum_j\sum_{i=1}^{\lambda^{\mathsf{t}}_j}(2i-1)=\sum_j(\lambda^{\mathsf{t}}_j)^2$.

Exercise 3.3. (a) $q^3+q^{-3}=(q+q^{-1})^3-3(q+q^{-1})=(y+2)^3-3(y+2)=y^3+6y^2+9y+2$. (b) $y^G=q^G+(\text{lower})+q^{-G}$, so subtracting c_Gy^G lowers the top degree while preserving the symmetry.

Exercise 4.1. $s_{(3)}=h_3=q^{3/2}/([1][2][3])$, and the hook formula gives $q^{6/4}/([3][2][1])$. $s_{(1,1,1)}=e_3=q^{-3/2}/([1][2][3])$, hook formula $q^{-6/4}/([3][2][1])$.

Exercise 4.2. The prefactor is $\frac{1+bz}{1+az} = 1 + (b-a)z - (b-a)az^2 + \dots$ with $a = q^{-1/2}$, $b = q^{1/2}$. Therefore $e_1 = e_1(q^\rho) + [1] = h_1(q^{(1)+\rho})$, and $e_2 = e_2(q^\rho) + [1]e_1(q^\rho) - [1]q^{-1/2} = \frac{q^{-1/2}}{[1][2]} + 1 - (1-q^{-1}) = q^{-1} + \frac{q^{-1/2}}{[1][2]}$.

Exercise 4.3. The e-form gives $e_1e_2 - e_3$ (since $(2,1)^{\mathsf{t}} = (2,1)$). From (12): $h_1 = e_1$, $h_2 = e_1^2 - e_2$, $h_3 = e_1^3 - 2e_1e_2 + e_3$, so $h_1h_2 - h_3 = e_1e_2 - e_3$.

Exercise 5.1. $C_{(2)(1)\varnothing} = \frac{1}{[1]}\big(q + \frac{q^{1/2}}{[1][2]}\big) = \frac{q}{[1]} + \frac{q^{1/2}}{[1]^2[2]} = \frac{q^{5/2} - q^{3/2} + q^{-1/2}}{[1]^2[2]}$, using $q[1][2] = q^{5/2} - q^{3/2} - q^{1/2} + q^{-1/2}$. Directly, $C_{\varnothing(2)(1)} = s_{(2)}(q^\rho)\, s_{(1)}(q^{(2)+\rho})$ and $h_1(q^{(2)+\rho}) = \frac{1}{[1]} + (q^{3/2} - q^{-1/2}) = \frac{1}{[1]} + q^{1/2}[2]$, so $C_{\varnothing(2)(1)} = \frac{q^{1/2}}{[1][2]}\big(\frac{1}{[1]} + q^{1/2}[2]\big)$, the same.

Exercise 5.2. Under $q \to 1/q$, $[h] \to -[h]$, so $s_\lambda(q^\rho) \to (-1)^{|\lambda|} q^{-\kappa(\lambda)/4} / \prod [h] = (-1)^{|\lambda|} s_{\lambda^{\mathsf{t}}}(q^\rho)$.

Exercise 5.3. $q \cdot \frac{q^{-1/2}}{[1][2]} = \frac{q^{1/2}}{[1][2]}$. Here $\kappa((2)) + \kappa(\varnothing) + \kappa(\varnothing) = 2$.

Exercise 7.1. Since $[1]^2[2]^2[3]^2 = y \cdot y(y+4) \cdot y(y+3)^2$, we have $Z_1 = -\frac{1}{y}$, $Z_2 = \frac{2}{y^2(y+4)}$, $Z_3 = -\frac{y+6}{y^3(y+4)(y+3)^2}$, and

$$F_3 = -\frac{y+6}{y^3(y+4)(y+3)^2} + \frac{2}{y^3(y+4)} - \frac{1}{3y^3} = \frac{-3(y+6) + 6(y+3)^2 - (y+4)(y+3)^2}{3y^3(y+4)(y+3)^2}. \tag{84}$$

The numerator is $-3y - 18 + (6y^2 + 36y + 54) - (y^3 + 10y^2 + 33y + 36) = -y^3 - 4y^2 = -y^2(y+4)$, so $F_3 = -\frac{1}{3y(y+3)^2} = -\frac{1}{3[3]^2}$.

Exercise 7.2. Z_B: $\lambda_1 = (1)$, sign $(-1)^{(0+1)} = -1$, vertices $C_{(1)\varnothing\varnothing}C_{\varnothing\varnothing(1)}$: $Z_B = -1/y$. $Z_F = 2/y$ (edges 2 and 4, sign $+1$). Z_{B+F}: $\lambda_3 = (1)$ gives $(-1)^3/y = -1/y$, $\lambda_1 = \lambda_2 = (1)$ gives $(-1) \cdot C_{(1)\varnothing(1)} \cdot \frac{1}{y} = -\frac{y+1}{y^2}$, and $\lambda_1 = \lambda_4 = (1)$ gives the same. Therefore $Z_{B+F} = -\frac{1}{y} - \frac{2(y+1)}{y^2}$ and $F_{B+F} = Z_{B+F} - Z_BZ_F = -\frac{1}{y} - \frac{2y+2}{y^2} + \frac{2}{y^2} = -\frac{3}{y}$, i.e. $n^0_{B+F} = 3$, and $F_B = -1/y$ gives $n^0_B = 1$.

Exercise 7.3. Each such term is $(-1)^3 q^{\kappa(\lambda)} s_{\lambda^{\mathsf{t}}}(q^\rho) s_\lambda(q^\rho) = -q^{\kappa(\lambda)} / \prod [h]^2$, so the 9 terms give $-3\Big(\frac{q^6 + q^{-6}}{[1]^2[2]^2[3]^2} + \frac{1}{[1]^4[3]^2}\Big)$.

References

[1] M. Aganagic, A. Klemm, M. Mariño, C. Vafa, *The topological vertex*, Commun. Math. Phys. 254 (2005) 425–478, `arXiv:hep-th/0305132`.

[2] Y. Konishi, *Integrality of Gopakumar-Vafa invariants of toric Calabi-Yau threefolds*, Publ. RIMS 42 (2006) 605–648, `arXiv:math/0504188`.

[3] P. Peng, *A simple proof of Gopakumar-Vafa conjecture for local toric Calabi-Yau manifolds*, Commun. Math. Phys. 276 (2007) 551–569, `arXiv:math/0410540`.

[4] M. Mariño, *Chern-Simons theory and topological strings*, Rev. Mod. Phys. 77 (2005) 675–720, `arXiv:hep-th/0406005`.

[5] A. Iqbal, A.-K. Kashani-Poor, *The vertex on a strip*, Adv. Theor. Math. Phys. 10 (2006) 317–343, arXiv:hep-th/0410174.

[6] A. Iqbal, A.-K. Kashani-Poor, *SU(N) geometries and topological string amplitudes*, Adv. Theor. Math. Phys. 10 (2006) 1–32, arXiv:hep-th/0306032.

[7] T.-M. Chiang, A. Klemm, S.-T. Yau, E. Zaslow, *Local mirror symmetry: calculations and interpretations*, Adv. Theor. Math. Phys. 3 (1999) 495–565, arXiv:hep-th/9903053.

[8] S. Katz, A. Klemm, C. Vafa, *M-theory, topological strings and spinning black holes*, Adv. Theor. Math. Phys. 3 (1999) 1445–1537, arXiv:hep-th/9910181.

[9] R. Gopakumar, C. Vafa, *M-theory and topological strings I, II*, arXiv:hep-th/9809187, arXiv:hep-th/9812127.

[10] A. Okounkov, N. Reshetikhin, C. Vafa, *Quantum Calabi-Yau and classical crystals*, in: The unity of mathematics, Progr. Math. 244, Birkhäuser 2006, arXiv:hep-th/0309208.

[11] J. Li, C.-C. M. Liu, K. Liu, J. Zhou, *A mathematical theory of the topological vertex*, Geom. Topol. 13 (2009) 527–621, arXiv:math/0408426.

[12] N. Nekrasov, *Seiberg-Witten prepotential from instanton counting*, Adv. Theor. Math. Phys. 7 (2003) 831–864, arXiv:hep-th/0206161.

[13] I. G. Macdonald, *Symmetric functions and Hall polynomials*, 2nd ed., Oxford University Press, 1995.

[14] J. A. De Loera, J. Rambau, F. Santos, *Triangulations: structures for algorithms and applications*, Springer, 2010.